

COMPASS: Robust Feature Conformal Prediction for Medical Segmentation Metrics

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Abstract

In clinical applications, the utility of segmentation models is often based on the accuracy of derived downstream metrics such as organ size, rather than by the pixel-level accuracy of the segmentation masks themselves. Thus, uncertainty quantification for such metrics is crucial for decision-making. Conformal prediction (CP) is a popular framework to derive such principled uncertainty guarantees, but applying CP naively to the final scalar metric is inefficient because it treats the complex, non-linear segmentation-to-metric pipeline as a black box. We introduce COMPASS, a practical framework that generates efficient, metric-based CP intervals for image segmentation models by leveraging the inductive biases of their underlying deep neural networks. COMPASS performs calibration directly in the model’s representation space by perturbing intermediate features along low-dimensional subspaces maximally sensitive to the target metric. We prove that COMPASS achieves valid marginal coverage under exchangeability and nestedness assumptions. Empirically, we demonstrate that COMPASS produces significantly tighter intervals than traditional CP baselines on four medical image segmentation tasks for area estimation of skin lesions and anatomical structures. Furthermore, we show that leveraging learned internal features to estimate importance weights allows COMPASS to also recover target coverage under covariate shifts. COMPASS paves the way for practical, metric-based uncertainty quantification for medical image segmentation.

1 Introduction

Uncertainty quantification is of critical need in medical image analysis, a field used for decision support in high-stakes clinical diagnosis and treatment planning applications [1, 2]. A fundamental task in medical image analysis is image segmentation, the task of separating anatomical structures and lesions from each other within an image. Deep learning models, particularly U-Net variants [3, 4], have achieved state-of-the-art performance in medical image segmentation. In practice, the outputs of these models (“segmentation maps”) are often treated as an intermediate result that are then used to automatically derive downstream metrics of interest (known as “radiomics”), such as the areas/volumes or texture patterns of specific anatomic regions (Figure 1, left). These derived metrics are then used for decision support to guide clinicians in diagnosis and treatment.

Conformal prediction (CP) has emerged as a popular, statistically principled uncertainty quantification framework of choice in machine learning, providing guarantees without restrictive distributional assumptions [5, 6, 7, 8, 9]. While well-studied in the context of typical prediction tasks involving scalar output variables, CP is less explored for tasks such as medical image segmentation, in which the output variables are images. Existing CP methods for segmentation typically focus on deriving bounds for *pixel-level errors* [10, 11, 12, 13], which, while useful for understanding variations of local segmentation contours, may yield meaningless or misaligned intervals for downstream derived metrics. On the other hand, a recent study shows that treating the segmentation-to-metric pipeline as a black box and performing CP directly on the output metric space yields intervals that are well-aligned to the metrics (by construction), but are also often inefficient (i.e., large) because the internal biases of the pipeline are not exploited in the vanilla CP formulation [14].

To achieve more efficient intervals, a promising direction is to leverage the powerful inductive biases of neural networks by performing CP on their intermediate representations. Feature Conformal Prediction (FCP) [15, 16, 17] shows that by working in a semantic feature space, it is possible to generate provably tighter prediction intervals. However, the FCP algorithm requires solving a complex optimization to find the closest adversarial feature vector for each data point, which is computationally prohibitive for high-dimensional feature spaces of typical CNN and transformer architectures which are the workhorses of modern medical image segmentation.

To bridge this gap, we introduce **Conformal Metric Perturbation Along Sensitive Subspaces (COMPASS)**, a framework to perform feature CP in a tractable manner to generate valid and efficient prediction intervals for any differentiable metric derived from the output of a neural network. The core concept behind COMPASS is to linearly perturb the features outputted by the network at a particular layer along data-specific directions that are highly sensitive to the metric of interest (Figure 1, center). To make this process tractable for typical neural network layers with a large number of features, we propose computing a low-dimensional manifold of any given layer by applying principal component analysis (PCA) on the gradients of the output metric with respect to each of the layer’s features. This manifold represents the principal directions of sensitivity of the output with respect to that layer. We prove that linear perturbations in the latent space achieve marginal coverage in the output metric space under exchangeability and nestedness assumptions, i.e., for a fresh test point, the ground truth will be contained within the prediction intervals generated by linear feature perturbations. Furthermore, we show that a simple weighted variant of COMPASS may be used to correct for covariate shifts.

We evaluated COMPASS on public datasets for four medical image segmentation tasks: colorectal cancer in histopathology images (EBHI) [18], skin lesion segmentation (HAM10000) [19], thyroid nodule segmentation in ultrasound images (TN3K) [20], and polyp segmentation in endoscopic images (Kvasir) [21]. Results show that while standard CP methods achieve valid coverage, they often produce unnecessarily wide prediction intervals. On the other hand, COMPASS finds semantically meaningful directions in the latent space that correspond to monotonic metric changes, resulting in efficient intervals. Furthermore, the weighted extension of COMPASS recovers target coverage under covariate shifts and is the most efficient across all weighted baseline methods. COMPASS paves the way for practical, metric-based uncertainty quantification for medical image segmentation.

2 Method

Let (X, Y) denote an input image and its corresponding ground-truth metric randomly drawn from an unknown distribution D . We consider a segmentation model that we can conceptually decompose into a feature extractor f and a subsequent downstream predictor g (see Figure 1-left). The function f maps the input image to a representation, $\hat{Z} = f(X)$, which can be the output of any intermediate layer of the network. The function g then maps this representation to the segmentation mask. Finally, a differentiable metric function h maps the final mask to a scalar prediction, $\hat{Y} = h(g(\hat{Z}))$. We use lowercase letters to denote the specific values of a particular realization (x, y) : $\hat{z} = f(x)$ and $\hat{y} = h(g(\hat{z}))$.

2.1 Theoretical Coverage under Linear Latent Perturbations

To provide rigorous uncertainty quantification for the metric Y , we consider *linear perturbations in latent space* along a direction $\Delta \in \mathcal{Z}$ and define intervals in the metric space. We first formalize the nestedness condition, which is a fundamental requirement for the validity of any conformal procedure [5, 6]. This condition guarantees that a larger perturbation magnitude yields a larger (or equal-sized) prediction set, which is necessary for the quantile-based coverage proof to hold. We explicitly state this assumption because, while trivially satisfied by construction in standard output-space methods, nestedness becomes a non-trivial condition in deep feature spaces with non-linear mappings.

Definition 1 (Nestedness). *Let $m_x : \mathbb{R} \rightarrow \mathbb{R}$ denote a real-valued function associated with each input $x \in \mathcal{X}$. For $\beta \geq 0$, define the interval*

$$S_\beta(x) := [m_x(-\beta), m_x(+\beta)].$$

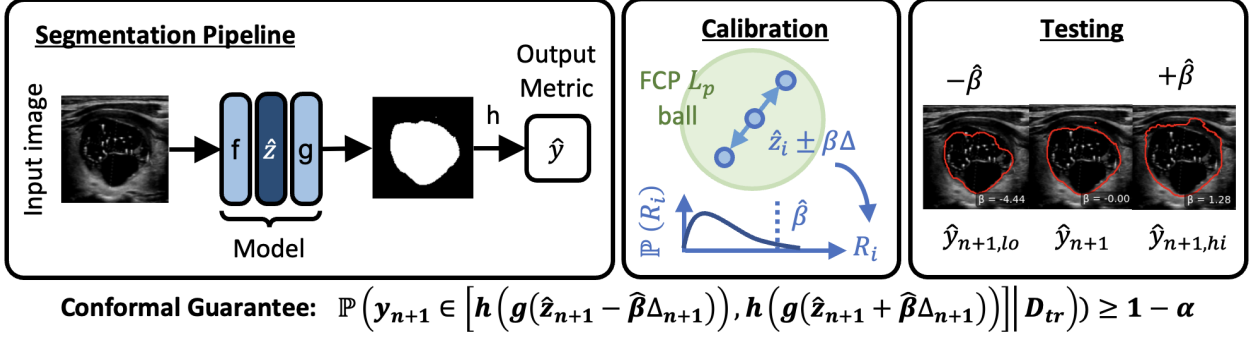


Figure 1: **Overview of COMPASS.** (Left) A medical image segmentation network predicts a segmentation “map” from an input image. We conceptually decompose this network into a function f which maps the image to latent features $\hat{z}$, and a function g that maps $\hat{z}$ to the output map. The map may then be used to compute a (differentiable) downstream metric $\hat{y}$ via the function h . (Center) We linearly perturb calibration features $\hat{z}_i$ in a sample-specific direction Δ_i to find the scores R_i . The scores are used to find the conformal quantile $\hat{\beta}$. (Right) At test time for subject $n + 1$, we perturb the features $\hat{z}_{n+1}$ in the direction Δ_{n+1} with magnitude $\hat{\beta}$. Under the assumptions of nestedness and exchangeability, the resulting prediction interval achieves marginal coverage (bottom).

The family $\{S_\beta(x)\}_{\beta \geq 0}$ is said to be nested for x if

$$\beta_1 \leq \beta_2 \Rightarrow S_{\beta_1}(x) \subseteq S_{\beta_2}(x).$$

Theorem 1 (Split-Conformal Coverage under Linear Latent Perturbations). *Let $(X_i, Y_i)_{i \geq 1}$ be exchangeable random pairs with $X_i \in \mathcal{X}$ and $Y_i \in \mathbb{R}$, and split the data into a training set D_{tr} and a calibration set $D_{cal} = \{(X_i, Y_i)\}_{i=1}^n$. Using D_{tr} , fit a segmentation model with decoder $g : \mathcal{Z} \rightarrow \mathcal{S}$, and let $\hat{z}(x) \in \mathcal{Z}$ denote the latent vector computed deterministically by the trained encoder for input x . Let $h : \mathcal{S} \rightarrow \mathbb{R}$ be a measurable, differentiable metric, and let $\Delta \in \mathcal{Z}$ be any measurable direction that depends only on D_{tr} .*

For $x \in \mathcal{X}$ and $\beta \geq 0$, define

$$m_x(\beta) := (h \circ g)(\hat{z}(x) + \beta\Delta), \quad S_\beta(x) := [m_x(-\beta), m_x(+\beta)].$$

Assume that for every x , the family $\{S_\beta(x)\}_{\beta \geq 0}$ is nested. For each calibration pair (X_i, Y_i) , define the non-conformity score

$$R_i := \inf\{\beta \geq 0 : Y_i \in S_\beta(X_i)\} \in [0, \infty],$$

and let $\hat{\beta}$ be the $\lceil (1 - \alpha)(n + 1) \rceil$ -th smallest value among $\{R_1, \dots, R_n\}$, where $\alpha \in (0, 1)$ is the user-specified mis-coverage level.

Then, for a fresh test pair (X_{n+1}, Y_{n+1}) , the prediction set $S_{\hat{\beta}}(X_{n+1})$ satisfies

$$\mathbb{P}(Y_{n+1} \in S_{\hat{\beta}}(X_{n+1}) \mid D_{tr}) \geq 1 - \alpha.$$

Proof sketch. Exchangeability of (X_i, Y_i) ensures that the rank of the test score R_{n+1} among $\{R_1, \dots, R_n, R_{n+1}\}$ is uniformly distributed; by construction of $\hat{\beta}$ and the nestedness assumption, the conformal guarantee holds. See Appendix A for full proof.

Intuition. Theorem 1 can be understood as an instance of conformal risk control [13], where the risk being controlled is the mis-coverage loss for a downstream metric, and the intervals generated by our perturbation scheme form the required nested family of prediction sets. Nestedness ensures that the sets $S_\beta(x)$ are weakly monotone with β . This property guarantees that increasing β cannot exclude a previously covered label Y , so R_i remains the minimal perturbation needed to include Y_i , analogous to residuals in standard CP. By exchangeability, the test score R_{n+1} is equally likely to occupy any rank among the calibration scores, and the quantile construction of $\hat{\beta}$ ensures the $1 - \alpha$ coverage.

2.2 COMPASS: Conformal Metric Perturbation Along Sensitive Subspaces

The proposed framework COMPASS is built on the insight that we can directly calibrate a downstream metric by perturbing a model’s representations along specific, data-dependent directions, Δ . As our theoretical guarantee (Theorem 1) applies to any measurable perturbation direction, the key to an efficient method lies in choosing a direction that is highly sensitive to the metric of interest. We explore two natural choices for this representation space: the model’s final output logits and a deeper internal feature layer.

The first and simplest approach (COMPASS-L) operates on the model’s final logits. A uniform, scalar shift applied to this tensor directly modulates the model’s overall confidence before the final activation function. This is equivalent to defining the sensitive direction, Δ_i , as a tensor of ones. A richer alternative is to perturb an internal feature representation $\hat{z}$, as done in FCP [15]. However, a naive search for an optimal perturbation in high-dimensional spaces is computationally intractable and sensitive to poor local minima. Thus, perturbation directions will not lead to monotonically increasing or decreasing output metrics. To overcome this, we propose a data-driven method to identify a low-dimensional sensitive subspace that is globally effective. We first compute Jacobians of the metric $\hat{y}$ with respect to $\hat{z}$ to provide a local, linear map of the metric’s sensitivity with respect to the features for each training sample:

$$J_i := \frac{dh(g(\hat{z}_i))}{d\hat{z}_i} \in \mathbb{R}^{C \times H \times W},$$

where $\hat{z}_i = f(x_i)$. Because the full spatial Jacobian is often too high-dimensional to be practical, we apply Principal Component Analysis (PCA) to the set of these vectors from the training set, $\{J_i\}_{i \in D_{\text{tr}}}$, and select the matrix of the top L eigenvectors $V_L \in \mathbb{R}^{C \times L}$. In the case of CNNs with image-shaped features, we first sum features per channel before applying PCA.

For any given sample, we find its particular sensitive direction $\mathbf{d}_i$ by projecting its sensitivity vector onto this learned subspace, and normalization it to produce direction vector Δ_i :

$$\mathbf{d}_i = V_L V_L^T J_i, \quad \Delta_i = \mathbf{d}_i / \|\mathbf{d}_i\|_2.$$

2.3 COMPASS Calibration and Inference

Regardless of the features used to compute direction Δ_i , the calibration and inference procedures for COMPASS follow the standard CP protocol. On the calibration set $\{(x_i, y_i)\}_{i \in D_{\text{cal}}}$, our goal is to find the non-conformity score R_i for each sample. We first define the prediction interval for a given perturbation magnitude $\beta \geq 0$ as:

$$S_\beta(x_i) := [h(g(\hat{z}_i - \beta \Delta_i)), h(g(\hat{z}_i + \beta \Delta_i))].$$

The score R_i is then the minimal magnitude that ensures coverage of the true value y_i :

$$R_i := \inf\{\beta \geq 0 : y_i \in S_\beta(x_i)\}.$$

To find this score efficiently, we perform a binary search with $\mathcal{O}(\log N)$ complexity. From the set of calibration scores $\{R_i\}_{i=1}^n$, we then compute the calibrated quantile $\hat{\beta}$ with the standard finite-sample correction:

$$\hat{\beta} := Q\left(\{R_i\}_{i=1}^n, \frac{\lceil (1 - \alpha)(n + 1) \rceil}{n}\right).$$

At inference time, for a new sample x_{n+1} , we compute its direction Δ_{n+1} and form the final prediction interval by applying the calibrated quantile $\hat{\beta}$:

$$S_{\hat{\beta}}(x_{n+1}) := [h(g(\hat{z}_{n+1} - \hat{\beta} \Delta_{n+1})), h(g(\hat{z}_{n+1} + \hat{\beta} \Delta_{n+1}))].$$

By Theorem 1, this interval is guaranteed to achieve the desired marginal coverage. For a full outline of the algorithm, including the asymmetric case, refer to Appendix C.

While symmetric perturbations are effective when the metric responds similarly to positive and negative perturbations, we often observe an asymmetric relationship when the metric responds differently in positive and negative directions. In such cases, an asymmetric calibration is necessary to construct a more adaptive and efficient interval. We achieve this by finding separate non-conformity scores for the upper and lower bounds. We discuss this asymmetric version of COMPASS and provide an equivalent algorithm in Appendix B.

2.4 Weighted COMPASS for Distribution Shifts

While Theorem 1 provides a robust guarantee of marginal coverage, its validity rests on the critical assumption that the calibration and test data are exchangeable. In many real-world applications, such as medical image segmentation, this assumption is frequently violated due to the variations in how the data is collected, processed, and interpreted. Under such distribution shifts, the unweighted quantile $\hat{\beta}$ is no longer guaranteed to provide the target coverage level on the test set, leading to systematic undercoverage and unreliable prediction intervals.

To address this limitation, we employ Weighted Conformal Prediction (WCP) [22, 23] to restore the coverage guarantee by re-weighting the calibration samples. The weight for a calibration sample X_i is ideally the density ratio $w(X_i) = p_{\text{test}}(X_i)/p_{\text{cal}}(X_i)$. In practice, this ratio is unknown and typically estimated by training an auxiliary classifier $\mathcal{A} : \mathcal{X} \rightarrow [0, 1]$ to distinguish between samples from the calibration and test sets. The predicted probability $\hat{p}(x) = \mathcal{A}(x)$ that a sample x belongs to the test set is then used to compute the weights, effectively adjusting the calibration procedure to account for the distribution shift. We now extend Theorem 1 to the weighted setting, providing a formal coverage guarantee under distribution shift. The setup remains identical.

Proposition 1 (Validity of Weighted COMPASS under Covariate Shift). *Let $D_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$ be n exchangeable pairs drawn from a distribution P_{cal} , and let (X_{n+1}, Y_{n+1}) be a fresh test pair from a potentially different distribution P_{test} . Let the non-conformity scores $R_i = R(X_i, Y_i)$ be computed as described in Theorem 1. Let the weights $w_i = w(X_i)$ be the true density ratio $p_{\text{test}}(X_i)/p_{\text{cal}}(X_i)$. Let $\hat{\beta}_w$ be the weighted $(1 - \alpha)$ -quantile of the calibration scores $\{R_1, \dots, R_n\}$ with corresponding weights $\{w_1, \dots, w_n\}$, defined as*

$$\hat{\beta}_w := \inf \left\{ \beta \geq 0 : \frac{\sum_{i=1}^n w_i \mathbf{1}\{R_i \leq \beta\}}{\sum_{j=1}^n w_j} \geq 1 - \alpha \right\}.$$

Then, for the fresh test pair $(X_{n+1}, Y_{n+1}) \sim P_{\text{test}}$, the prediction set $S_{\hat{\beta}_w}(X_{n+1})$ satisfies

$$\mathbb{P}(Y_{n+1} \in S_{\hat{\beta}_w}(X_{n+1}) \mid D_{\text{tr}}, D_{\text{cal}}) \geq 1 - \alpha.$$

Proof Sketch. This is a direct application of WCP [22]. The validity of this framework applies to any valid non-conformity score. Our scores are valid as they are from a deterministic function of the samples and the pre-trained model. When the weights represent the true density ratio, the weighted empirical distribution of the calibration scores is an unbiased estimator of the test score distribution. Thus, the coverage guarantee from the original theorem directly applies. $\square$

In practice, the true oracle is not available, and the coverage guarantee holds *approximately*, with the quality of the approximation depending on the accuracy of the density ratio estimates. To correct for the induced distribution shift, we explore three correction strategies: 1) Class: available ground truth class labels as features and weights are computed directly from the known class prevalences, 2) Latent: Model’s latent representations summed on the spatial dimension as features [24], and 3) Jacobian: model’s internal geometric sensitivity (jacobians) summed as features.

3 Experiments

We evaluated COMPASS across four medical image segmentation tasks: 1) segmentation on H&E histopathology images from the EBHI dataset (H&E) [18], 2) skin lesion segmentation on dermoscopic images from the HAM10000 dataset (Skin Lesion) [19], 3) thyroid nodule segmentation from the TN3K dataset (Nodule) [20], and 4) gastrointestinal polyp segmentation (PolyP) [21]. We trained all models using the standard U-Net architecture from MONAI [25, 26]. We focused on segmented object size (area) as the downstream clinical metric of interest. Object size is among the most widely adopted quantitative biomarkers across diverse clinical applications [27, 28]. We repeated each experiment over 100 randomized splits to compute average coverage and interval sizes. We provide details on architecture, preprocessing, training, calibration, and testing details in Appendix E. We considered three types of calibration strategies:

1. Output-space calibration: These include split conformal prediction (SCP) [29], Conformalized Quantile Regression (Output-CQR) [30], and locally adaptive conformal prediction (Local CP) [29, 31, 32].

2. End-to-end calibration: We use Conformalized Quantile Regression (E2E-CQR) [24] on a model trained to directly produce pixel-wise lower and upper bounds using the Tversky loss [33].

3. Feature-space calibration: These include COMPASS-L, which directly calibrates the model’s final pre-activation outputs (logits), and COMPASS-J, which constructs prediction intervals by perturbing latent features along dominant directions. We omitted the original FCP method [15] from our main empirical comparison because we found its core adversarial search procedure to find non-conformity scores to be computationally intractable for large feature spaces and fails to reliably converge. We instead offer a comparison against a conceptual oracle benchmark for FCP in Appendix D: because FCP prediction set is defined by an L_p ball in the latent space [15] with bounds computed using Linear Relaxation based Perturbation Analysis (LiRPA) [34], we empirically find the minimal radius of the L_p ball, which, when propagated through the decoding function using LiRPA, achieves the target coverage for the final metric.

3.1 Standard Conformal Prediction

First, we qualitatively find that COMPASS finds semantically meaningful directions in the latent space that correspond to monotonic changes in metric (Figure 2) for all applications. Next, we compare interval size across methods and datasets for $\alpha = \{0.15, 0.1, 0.05\}$ (Table 1). We find that output space methods (SCP, CQR, Local) generally achieve valid coverage (Table 2 in Appendix D) but yield longer intervals. End-to-end CQR provides tighter intervals. COMPASS (-J and -L) achieves valid coverage with a significantly reduced interval size compared to E2E-CQR. Moreover, we observe an efficiency benefit of using deeper representations (COMPASS-J) compared to logits (COMPASS-L). COMPASS-J generally achieves the shortest interval lengths overall. This highlights the efficiency of COMPASS methods.

3.2 Weighted Conformal Prediction

To empirically validate the effectiveness of WCP and compare different weighting strategies, we design an experiment with a controlled, adversarial label shift for datasets that include class labels, namely H&E and Skin Lesion. Using the same architecture and training setup as the unweighted case, we create two scenarios where unweighted methods are expected to fail, thereby highlighting the ability of WCP to restore the target coverage and efficient interval lengths. The training set is the same as the unweighted case. The calibration set is constructed to be 1) “easy” by undersampling a difficult class and 2) “hard” by undersampling the easy class. The test set is constructed to be 1) “hard” by oversampling a difficult class and 2) easy by

Table 1: **COMPASS achieves more efficient interval sizes compared to baseline methods across different target coverages.** For 4 datasets and 100 random splits, we compare interval lengths at target coverages at $\alpha = \{0.15, 0.1, 0.05\}$. We show the output space, end-to-end, and feature calibration methods in red, blue, and green. The shortest mean interval lengths are bolded. For empirical coverages, see Appendix D.

Dataset	α	Interval Size (Mean $\pm$ STD, Pixels ²)					
		COMPASS-J	COMPASS-L	E2E-CQR	Local	Output-CQR	SCP
H&E	0.05	4635 $\pm$ 569	4542$\pm$449	5161 $\pm$ 465	6211 $\pm$ 598	5392 $\pm$ 338	5805 $\pm$ 732
	0.1	2512$\pm$160	2655 $\pm$ 205	2929 $\pm$ 312	3764 $\pm$ 322	3637 $\pm$ 237	3467 $\pm$ 265
	0.15	2364$\pm$145	2513 $\pm$ 145	2652 $\pm$ 169	3182 $\pm$ 249	3082 $\pm$ 284	2733 $\pm$ 199
Skin Lesion	0.05	1590$\pm$61	1633 $\pm$ 84	2570 $\pm$ 211	3953 $\pm$ 204	10936 $\pm$ 55	3296 $\pm$ 236
	0.1	1149$\pm$41	1170 $\pm$ 47	1344 $\pm$ 80	2580 $\pm$ 166	4911 $\pm$ 41	1828 $\pm$ 138
	0.15	920$\pm$24	932 $\pm$ 26	935 $\pm$ 40	1866 $\pm$ 62	2470 $\pm$ 36	1118 $\pm$ 77
Nodules	0.05	3184$\pm$190	3293 $\pm$ 208	4075 $\pm$ 320	3810 $\pm$ 234	7560 $\pm$ 49	4616 $\pm$ 375
	0.1	2388$\pm$130	2469 $\pm$ 141	2810 $\pm$ 184	3223 $\pm$ 99	5450 $\pm$ 48	3095 $\pm$ 221
	0.15	1940$\pm$106	2005 $\pm$ 138	2197 $\pm$ 181	2706 $\pm$ 148	4066 $\pm$ 88	2381 $\pm$ 154
PolyP	0.05	5061$\pm$497	5358 $\pm$ 563	8934 $\pm$ 843	12071 $\pm$ 2197	7618 $\pm$ 820	8305 $\pm$ 574
	0.1	3940$\pm$262	4101 $\pm$ 286	6329 $\pm$ 584	6324 $\pm$ 926	4887 $\pm$ 459	6194 $\pm$ 550
	0.15	3228$\pm$191	3414 $\pm$ 254	4593 $\pm$ 496	4391 $\pm$ 410	3951 $\pm$ 295	4533 $\pm$ 405

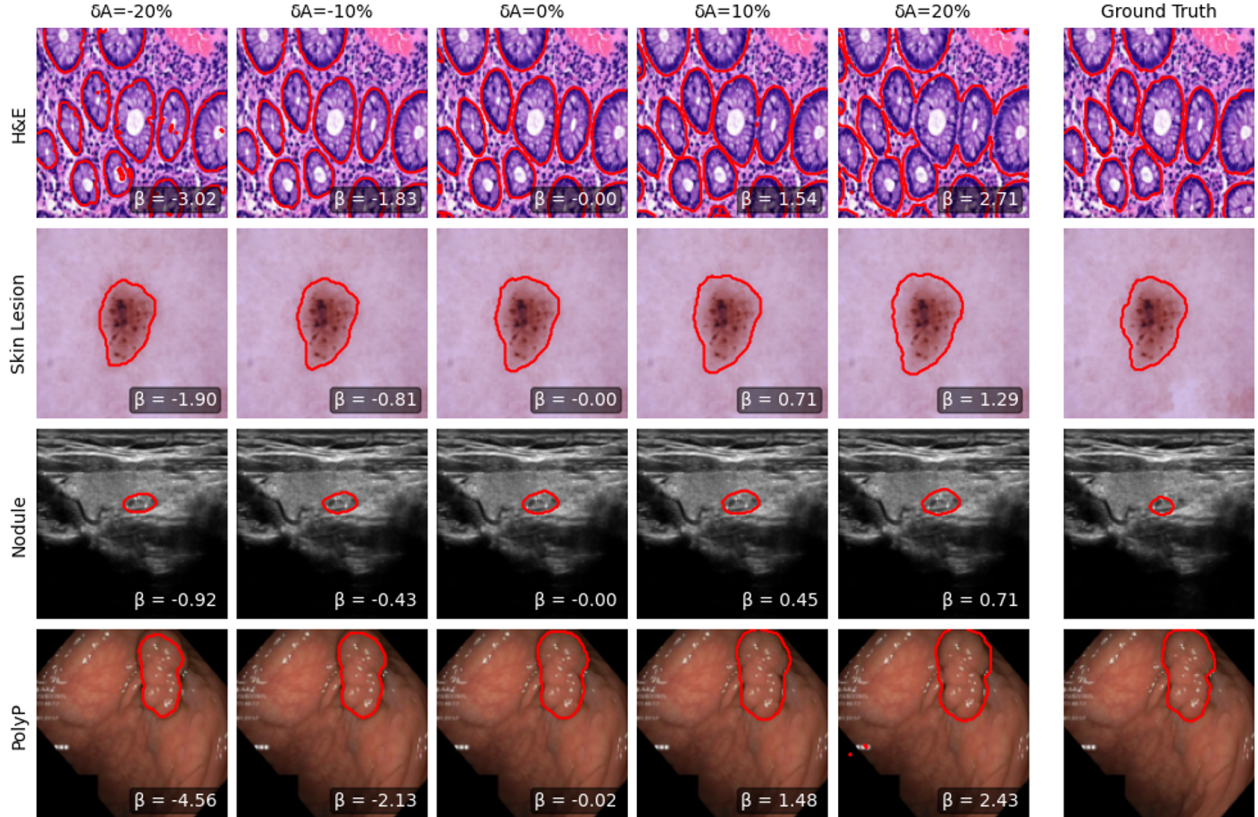


Figure 2: **As latent features are shifted along the perturbation directions Δ , the induced segmentation volumes monotonically expand.** Using COMPASS-J and for a sample from each dataset, we show a grid of segmentation contours (in red) with perturbation magnitudes β targeted at -20%, -10%, 0% (original prediction), +10%, +20% change in area (δA). We provide more examples in Appendix G.

undersampling the difficult class. For H&E, we place 60% and 40% of the Adenocarcinoma (easy) samples into the calibration and test sets. For Skin Lesion, we place 70% and 30% of the melanocytic nevi (hard) samples into the calibration and test sets. See Appendix E for the calibration and testing class distributions.

For output-space calibration methods, we used weights based on the ground truth class labels, which we consider as approximately “oracle” weights. Note that the theoretically perfect oracle weights are defined by the density ratio, which precisely corrects for the change in the full, high-dimensional distribution of features between the calibration and test sets. For end-to-end and feature space calibration, we used the latent features [24, 35, 36] and Jacobians to train auxiliary classifiers using gradient boosting machines [37] to distinguish between calibration and test sets. For end-to-end, we do not use Jacobian weights consistent with prior work [24].

We plot the distribution of interval size and mean coverage for unweighted, oracle, feature, and Jacobian weighted methods with $\alpha = 0.1$ (Figure 3). As expected, unweighted methods do not match the target coverage. The coverages for H&E and Skin Lesion tend to be under or overcovered, which is expected from the adversarial split procedure. We find that using class, feature, and Jacobian weightings allows target coverage recovery. In each weighting method that achieves target coverage, COMPASS (specifically COMPASS-J) achieves the shortest interval lengths. The slight undercoverage for feature-weighted E2E-CQR for H&E and class weighted methods in Skin Lesion can be attributed to the unavailability of the true oracle weights. In spite of this, COMPASS methods are the closest to target coverage. These results highlight the strength of COMPASS for correcting for covariate shifts. We show a table of results in Appendix D.

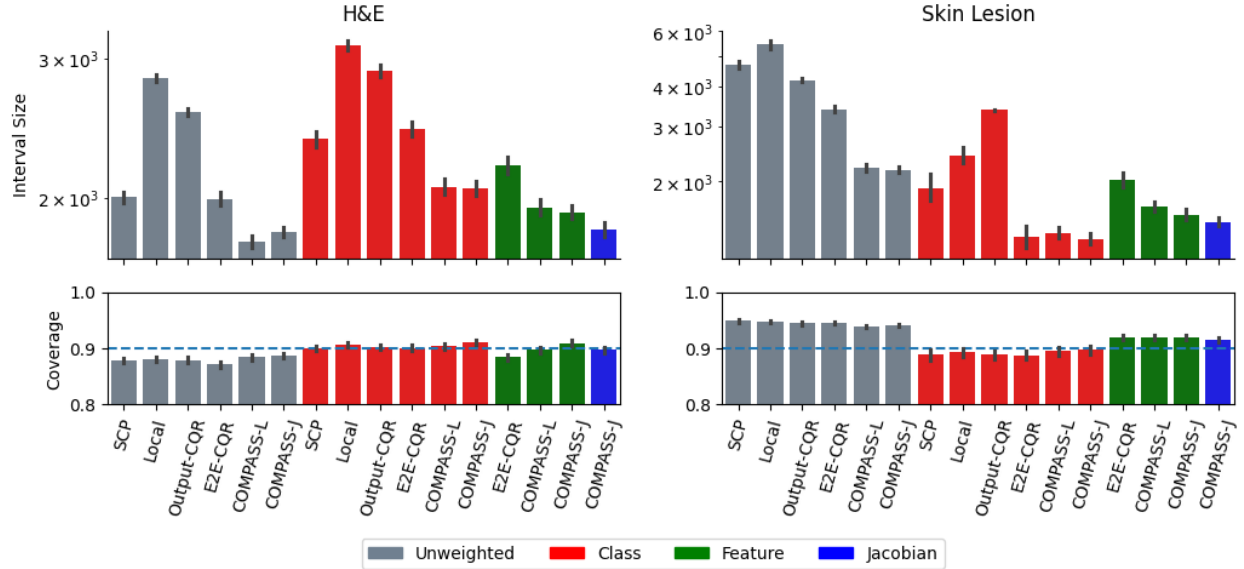


Figure 3: **COMPASS achieves the most efficient interval sizes under covariate shifts.** We show results for two datasets and compare weighting methods for 100 adversarial splits that maintain the same covariate shift for $\alpha = 0.1$. For H&E, we increased the proportion of “hard” samples in the test set. For Skin Lesion, we decreased the proportion of “hard” samples in the test set. We find that COMPASS methods achieve valid coverage and the most efficient intervals in each weighting method. We show the 95% confidence intervals.

3.3 Empirical Analysis of Statistical Efficiency

Unlike the original FCP framework, which provides a formal inequality under mild assumptions due to its reliance on the simple and convex $\mathcal{L}_p$ norm as a score, a similar theorem for COMPASS is intractable, because the COMPASS score $R_{COMPASS}$ is defined implicitly through a search process that depends on a highly non-linear and non-convex segmentation-to-metric pipeline. Instead, we investigated the reason for efficiency gains of COMPASS compared to output-space methods by finding the relationship between COMPASS scores and SCP scores R_{SCP} .

The statistical efficiency of COMPASS is a direct consequence of a compressive power-law relationship between feature-space scores ($R_{COMPASS}$) and output-space errors (R_{SCP}). A log-log plot of these quantities reveals a linear relationship with a scaling exponent slope < 1 (Figure 4, top). This sub-linear scaling is the direct mechanism for the tail-end compression of the score distribution (Figure 4, bottom). Thus, the largest output-space errors are systematically mapped to disproportionately smaller feature-space scores. A distribution with a compressed tail necessarily has a smaller quantile, which is the fundamental mechanism that enables COMPASS to produce tighter, more statistically efficient prediction intervals.

4 Discussion

We introduced COMPASS, a unified framework to generate efficient, metric-based prediction intervals by leveraging the inductive biases of neural networks. Under the assumption of nestedness and exchangeability, COMPASS creates practical feature-space calibration by perturbing a model’s intermediate representations along low-dimensional subspaces maximally sensitive to the target metric. Across four medical segmentation tasks for area estimation, COMPASS achieves valid coverage while producing significantly tighter intervals than traditional output-space and end-to-end baselines for both standard and weighted CP. We discuss several further points below.

COMPASS has several attractive properties. First, **it produces instance-adaptive intervals.** For COMPASS-J, the perturbation direction for any new sample is found by projecting its Jacobian onto the

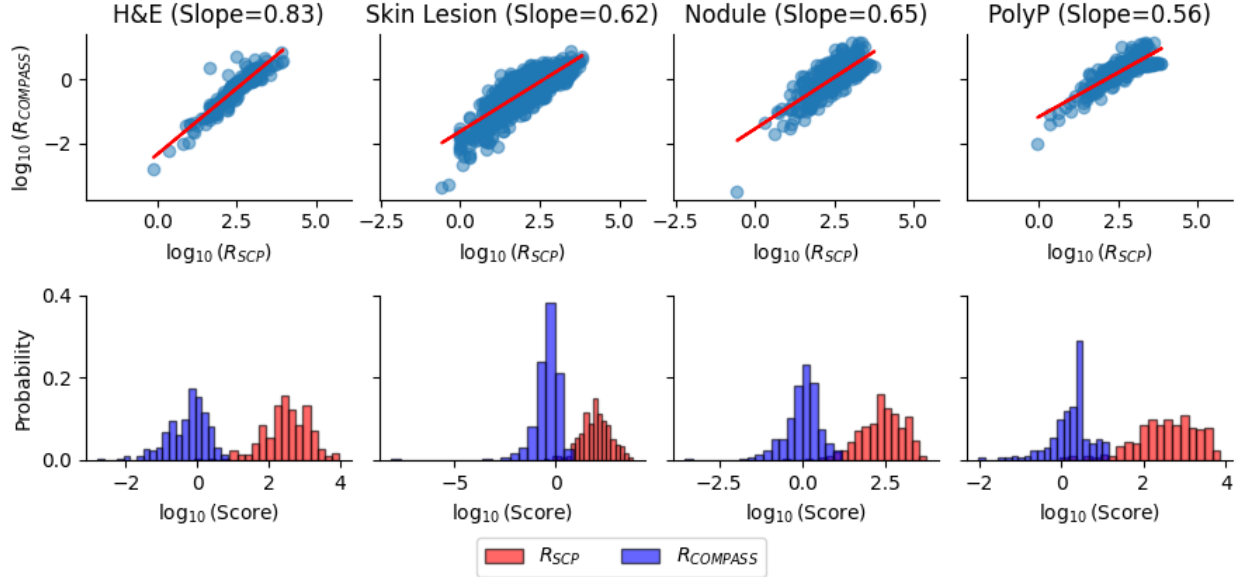


Figure 4: **The statistical efficiency of COMPASS is driven by a compressive power-law relationship between latent $R_{COMPASS}$ and output R_{SCP} space scores.** As R_{SCP} increases, the required latent space perturbation magnitude increases, but at a progressively slower rate since the scaling exponent (slope) is < 1 (Figure 4, top). This concave and sub-linear scaling is the direct cause of a tail-end compression of the score distribution (bottom). Thus, the long-tail errors are systematically transformed to much smaller feature-space scores.

principal component component and reconstructing the direction. Second, **COMPASS performs better than naive output-space calibration.** In tasks like segmentation, the final metric is a complex non-linear function of the model’s output (the pixel mask). Simply adding a margin to a final object size prediction is a crude approximation that fails to capture how the object size is actually derived from the underlying segmentation. In contrast, COMPASS exploits the model’s internal structure and spatial understanding to directly manipulate and calibrate the metric. COMPASS has more information and degrees of freedom, which allows it to achieve more efficient interval sizes. Third, **COMPASS performs better than methods trained using pixel-level losses** such as E2E-CQR. Such approaches optimize a *proxy* (pixel quantiles) for the downstream metric, that may or may not translate into accurate intervals for the final metric. On the other hand, COMPASS is directly calibrated to the final metric, and is therefore more applicable to practical clinical use cases.

Finally, **COMPASS is naturally applicable to various segmentation architectures.** We also present the interval lengths and coverage for SegResNet [38] in Appendix D. COMPASS produces the most efficient interval lengths on the majority of datasets across $\alpha = \{0.05, 0.1, 0.15\}$ for SegResNet.

Limitations. COMPASS’s performance is fundamentally dependent on the quality of the pre-trained model’s representations. Our framework presupposes that the model has learned an inductive bias where the chosen feature space has a reasonably monotonic and sensitive control over the downstream metric. If the features are poorly aligned with the metric, the resulting perturbation direction may be inefficient or non-monotonic, leading to wide or invalid intervals. Furthermore, as mentioned in prior work [24], WCP with feature or jacobian weights, is effective for moderate distribution shifts where the calibration and test distributions have significant overlap in the feature space. However, for large shifts, the estimated weights may be inaccurate, which can compromise the validity of the final interval.

5 Conclusion

We introduce COMPASS, a unified framework to generate efficient, metric-based prediction intervals by leveraging the inductive biases of neural networks. Under the assumption of nestedness, COMPASS creates practical feature-space calibration by perturbing a model’s intermediate representations along low-dimensional subspaces maximally sensitive to the target metric. Across four medical segmentation tasks for area estimation, COMPASS achieves valid coverage while producing significantly tighter intervals than traditional output-space and end-to-end baselines for both standard and weighted CP.

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Code Availability Code will be released at <https://github.com/matthewyccheung/compass> upon acceptance.

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A Proof of Theorem 1

Lemma 1 (Nestedness implies valid scores). *For any $x \in \mathcal{X}$, if $\{S_\beta(x)\}_{\beta \geq 0}$ is nested, then the nonconformity score*

$$R(x, y) := \inf\{\beta \geq 0 : y \in S_\beta(x)\}$$

is well-defined in $[0, \infty]$ and satisfies

$$y \in S_\beta(x) \iff R(x, y) \leq \beta.$$

By nestedness, the sets $S_\beta(x)$ expand monotonically in β . Thus, if y is contained in some $S_{\beta'}(x)$, then it is also contained in $S_\beta(x)$ for all $\beta \geq \beta'$. Hence the infimum in the definition of $R(x, y)$ exists. Moreover, $y \in S_\beta(x)$ if and only if β exceeds this infimum.

Lemma 2 (Exchangeability yields uniform ranks). *Let $(X_i, Y_i)_{i=1}^{n+1}$ be exchangeable, and let $R_i := R(X_i, Y_i)$. Then the rank of R_{n+1} among $\{R_1, \dots, R_n, R_{n+1}\}$ is uniformly distributed on $\{1, \dots, n+1\}$, conditional on D_{tr} .*

Since the nonconformity score $R(\cdot, \cdot)$ is a measurable, deterministic function of (X_i, Y_i) and D_{tr} , the exchangeability of (X_i, Y_i) implies exchangeability of $(R_i)_{i=1}^{n+1}$. Therefore the rank of R_{n+1} among all $n+1$ scores is uniform.

Theorem 2 (Split-conformal coverage). *Under the assumptions of Theorem 1, for any $\alpha \in (0, 1)$, the prediction set $S_{\hat{\beta}}(X_{n+1})$ satisfies*

$$\mathbb{P}(Y_{n+1} \in S_{\hat{\beta}}(X_{n+1}) \mid D_{\text{tr}}) \geq 1 - \alpha.$$

By Lemma 1, $Y_{n+1} \in S_\beta(X_{n+1})$ if and only if $R_{n+1} \leq \beta$. Hence

$$Y_{n+1} \in S_{\hat{\beta}}(X_{n+1}) \iff R_{n+1} \leq \hat{\beta}.$$

By Lemma 2, the rank of R_{n+1} is uniform among $\{1, \dots, n+1\}$. Since $\hat{\beta}$ is the $\lceil (1 - \alpha)(n+1) \rceil$ -th order statistic of $\{R_1, \dots, R_n\}$, it follows that

$$\mathbb{P}(R_{n+1} \leq \hat{\beta} \mid D_{\text{tr}}) \geq 1 - \alpha,$$

and the claim follows.

Theorem 1 generalizes the standard split conformal method. In the classical setting, $S_\beta(x) = [-\beta, +\beta]$ corresponds to symmetric absolute-error balls around a predictor. Here, the sets $S_\beta(x)$ are generated by perturbations of latent features through Δ and evaluation under the downstream metric $h \circ g$. The proof requires only nestedness and exchangeability, so the same rank-uniformity argument ensures coverage. Thus, standard split conformal prediction is recovered as a special case of the present framework.

B Asymmetric Perturbations

We extend the symmetric method to provide more flexible and efficient prediction intervals to the asymmetric case. This approach is particularly valuable when the relationship between latent perturbations and the downstream metric is non-linear or asymmetric. For instance, a positive perturbation of a certain magnitude might cause a large increase in volume, while a negative perturbation of the same magnitude results in only a small decrease. It also allows for independent control over the rate of under- and over-estimation errors, specified by α_{lo} and α_{hi} . To account for this, we calculate two distinct non-conformity scores $(R_{\text{lo}}, R_{\text{hi}})$ and two corresponding perturbation magnitudes $(\beta_{\text{lo}}, \beta_{\text{hi}})$. By calibrating the upper and lower bounds independently, Asymmetric COMPASS constructs an interval that adapts to these skewed error distributions, often resulting in tighter bounds than its symmetric counterpart in such cases.

Theorem 3 (Asymmetric Split-Conformal Coverage). *Let the setup be the same as in the original theorem, and assume the family of sets $\{S_\beta(x)\}_{\beta \geq 0}$ is nested for every x as per Definition 1.*

For each calibration pair (X_i, Y_i) , define two non-conformity scores:

$$\begin{aligned} R_{i,\text{lo}} &:= \inf\{\beta \geq 0 : m_{X_i}(-\beta) \leq Y_i\}, \\ R_{i,\text{hi}} &:= \inf\{\beta \geq 0 : m_{X_i}(+\beta) \geq Y_i\}. \end{aligned}$$

Let $\alpha_{\text{lo}}, \alpha_{\text{hi}} \in (0, 1)$ be user-specified miscoverage rates. Let $\hat{\beta}_{\text{lo}}$ be the $\lceil (1 - \alpha_{\text{lo}})(n + 1) \rceil$ -th smallest value among $\{R_{1,\text{lo}}, \dots, R_{n,\text{lo}}\}$, and let $\hat{\beta}_{\text{hi}}$ be the $\lceil (1 - \alpha_{\text{hi}})(n + 1) \rceil$ -th smallest value among $\{R_{1,\text{hi}}, \dots, R_{n,\text{hi}}\}$.

Then, for a fresh test pair (X_{n+1}, Y_{n+1}) , the prediction set

$$S(X_{n+1}) := [m_{X_{n+1}}(-\hat{\beta}_{\text{lo}}), m_{X_{n+1}}(+\hat{\beta}_{\text{hi}})]$$

satisfies the marginal coverage guarantee

$$\mathbb{P}(Y_{n+1} \in S(X_{n+1}) \mid D_{\text{tr}}) \geq 1 - (\alpha_{\text{lo}} + \alpha_{\text{hi}}).$$

B.1 Proof of Theorem 3

Lemma 3 (Nestedness implies valid one-sided scores). *For any $x \in \mathcal{X}$, if $\{S_\beta(x)\}_{\beta \geq 0}$ is nested, then the nonconformity scores $R_{\text{lo}}(x, y)$ and $R_{\text{hi}}(x, y)$ are well-defined in $[0, \infty]$ and satisfy*

$$\begin{aligned} m_x(-\beta) \leq y &\iff R_{\text{lo}}(x, y) \leq \beta, \\ m_x(+\beta) \geq y &\iff R_{\text{hi}}(x, y) \leq \beta. \end{aligned}$$

Nestedness implies that the lower bound $m_x(-\beta)$ is non-increasing in β and the upper bound $m_x(+\beta)$ is non-decreasing in β . This one-sided monotonicity ensures that if a bound is satisfied for some β' , it remains satisfied for all $\beta \geq \beta'$. Hence, the infima in the definitions of the scores exist, and the equivalences hold.

Lemma 4 (Exchangeability yields uniform ranks). *Let $(X_i, Y_i)_{i=1}^{n+1}$ be exchangeable. Let $R_{i,\text{lo}} := R_{\text{lo}}(X_i, Y_i)$ and $R_{i,\text{hi}} := R_{\text{hi}}(X_i, Y_i)$. Then the ranks of $R_{n+1,\text{lo}}$ among $\{R_{1,\text{lo}}, \dots, R_{n+1,\text{lo}}\}$ and $R_{n+1,\text{hi}}$ among $\{R_{1,\text{hi}}, \dots, R_{n+1,\text{hi}}\}$ are each uniformly distributed on $\{1, \dots, n + 1\}$, conditional on D_{tr} .*

Since the nonconformity scores are measurable, deterministic functions of (X_i, Y_i) and D_{tr} , the exchangeability of the data pairs implies the exchangeability of each sequence of scores, $(R_{i,\text{lo}})_{i=1}^{n+1}$ and $(R_{i,\text{hi}})_{i=1}^{n+1}$. Therefore, the rank of the test score within each sequence is uniform.

Proof of Theorem 3. The total miscoverage event is the union of two one-sided miscoverage events. By the union bound,

$$\mathbb{P}(Y_{n+1} \notin S(X_{n+1})) \leq \mathbb{P}(Y_{n+1} < m_{X_{n+1}}(-\hat{\beta}_{\text{lo}})) + \mathbb{P}(Y_{n+1} > m_{X_{n+1}}(+\hat{\beta}_{\text{hi}})).$$

By Lemma 3, these one-sided events are equivalent to events concerning the scores:

$$\mathbb{P}(Y_{n+1} \notin S(X_{n+1})) \leq \mathbb{P}(R_{n+1,\text{lo}} > \hat{\beta}_{\text{lo}}) + \mathbb{P}(R_{n+1,\text{hi}} > \hat{\beta}_{\text{hi}}).$$

By Lemma 4 and the construction of $\hat{\beta}_{\text{lo}}$ and $\hat{\beta}_{\text{hi}}$ as quantiles, the probabilities of these rank-based events are bounded by α_{lo} and α_{hi} , respectively. Thus,

$$\mathbb{P}(Y_{n+1} \notin S(X_{n+1})) \leq \alpha_{\text{lo}} + \alpha_{\text{hi}}.$$

The claim for the coverage probability, $\mathbb{P}(Y_{n+1} \in S(X_{n+1})) \geq 1 - (\alpha_{\text{lo}} + \alpha_{\text{hi}})$, follows. $\square$

This theorem provides a stronger guarantee than its symmetric counterpart. The proof relies on the same core principles of nestedness and exchangeability, but applies them to each bound independently before combining them with the union bound.

C Algorithms

In the COMPASS framework, we decompose the model into an encoder f that maps an input image x to a representation $\hat{z}$, and a decoder g that maps $\hat{z}$ to the final logits. A downstream metric function h then maps the resulting segmentation to a scalar value. For **COMPASS-L**, the representation $\hat{z}$ is the final logit tensor itself; the encoder f is the entire network and the decoder g is the identity function. For **COMPASS-J**, the representation $\hat{z}$ is an internal feature map; the encoder f is the part of the network up to that layer, and the decoder g is the remainder of the network. For clarity in the algorithms, we define the perturbation and metric evaluation function $P(\hat{z}, \Delta, \beta)$ as the interval $[h(g(\hat{z} - \beta\Delta)), h(g(\hat{z} + \beta\Delta))]$.

C.1 Algorithm 1: COMPASS

With the COMPASS algorithm, we construct a valid prediction interval for a downstream metric. The interval is *symmetric* in the representation space, meaning we perturb the representation by the same magnitude, $\pm\hat{\beta}$, along a data-specific direction, Δ . Following conventional CP procedures, we separate the algorithm into three stages: Training, Calibration, and Inference.

Algorithm 1 COMPASS

Require: Training D_{tr} , Calibration $D_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$, Test x_{n+1} , α , L .

Ensure: Prediction interval $S_{\hat{\beta}}(x_{n+1})$.

```

1: procedure TRAINING( $D_{\text{tr}}, L$ )
2:   Train the segmentation model  $(f, g) \leftarrow \arg \min_{f, g} \mathcal{L}_{\text{seg}}(D_{\text{tr}})$ .
3:   if using COMPASS-J then
4:      $\mathcal{J} \leftarrow \{\sum_{\text{spatial}} \nabla_{\hat{z}_j} h(g(f(x_j)))\}_{j \in D_{\text{tr}}}$ .
5:      $V_L \leftarrow \text{PCA}(\mathcal{J}, L)$ .
6:     return  $f, g, V_L$ .
7:   else ▷ COMPASS-L requires no extra training
8:     return  $f, g$ .
9:   end if
10: end procedure

11: procedure CALIBRATION( $D_{\text{cal}}, \text{model}, h, \alpha, \dots$ )
12:   Initialize scores  $\mathcal{R} \leftarrow \emptyset$ .
13:   for  $i = 1, \dots, n$  do
14:     Define representation  $\hat{z}_i$  and direction  $\Delta_i$ .
15:      $R_i \leftarrow \text{Symmetric-Binary-Search}(\hat{z}_i, \Delta_i, y_i)$ .
16:      $\mathcal{R} \leftarrow \mathcal{R} \cup \{R_i\}$ .
17:   end for
18:    $\hat{\beta} \leftarrow \text{Quantile}(\mathcal{R}, \frac{\lceil (1-\alpha)(n+1) \rceil}{n})$ .
19:   return  $\hat{\beta}$ 
20: end procedure

21: procedure INFERENCE( $x_{n+1}, \text{model}, h, \hat{\beta}, \dots$ )
22:   Define representation  $\hat{z}_{n+1}$  and direction  $\Delta_{n+1}$ .
23:    $S_{\hat{\beta}}(x_{n+1}) \leftarrow P(\hat{z}_{n+1}, \Delta_{n+1}, \hat{\beta})$ .
24:   return  $S_{\hat{\beta}}(x_{n+1})$ 
25: end procedure

```

C.2 Algorithm 2: Asymmetric Perturbations with COMPASS

We extend COMPASS to provide separate control over the lower and upper bounds of the prediction interval—the *asymmetric* case. We calculate two distinct non-conformity scores ($R_{\text{lo}}, R_{\text{hi}}$) and two corresponding perturbation magnitudes ($\hat{\beta}_{\text{lo}}, \hat{\beta}_{\text{hi}}$) based on separate miscoverage rates ($\alpha_{\text{lo}}, \alpha_{\text{hi}}$).

Algorithm 2 Asymmetric COMPASS

Require: Training D_{tr} , Calibration $D_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$, Test x_{n+1} , $\alpha_{\text{lo}}, \alpha_{\text{hi}}$, L .

Ensure: Prediction interval $S(x_{n+1})$.

```
1: procedure TRAINING( $D_{\text{tr}}, L$ ) ▷ Identical to Symmetric COMPASS
2:   This stage is identical to the Training procedure in Algorithm 1.
3: end procedure

4: procedure CALIBRATION( $D_{\text{cal}}, \text{model}, h, \alpha_{\text{lo}}, \alpha_{\text{hi}}, \dots$ )
5:   Initialize  $\mathcal{R}_{\text{lo}} \leftarrow \emptyset, \mathcal{R}_{\text{hi}} \leftarrow \emptyset$ .
6:   for  $i = 1, \dots, n$  do
7:     Define representation  $\hat{z}_i$  and direction  $\Delta_i$ .
8:      $(R_{i,\text{lo}}, R_{i,\text{hi}}) \leftarrow \text{Asymmetric-Binary-Search}(\hat{z}_i, \Delta_i, y_i)$ .
9:      $\mathcal{R}_{\text{lo}} \leftarrow \mathcal{R}_{\text{lo}} \cup \{R_{i,\text{lo}}\}, \mathcal{R}_{\text{hi}} \leftarrow \mathcal{R}_{\text{hi}} \cup \{R_{i,\text{hi}}\}$ .
10:  end for
11:   $\hat{\beta}_{\text{lo}} \leftarrow \text{Quantile}(\mathcal{R}_{\text{lo}}, \frac{\lceil (1-\alpha_{\text{lo}})(n+1) \rceil}{n})$ .
12:   $\hat{\beta}_{\text{hi}} \leftarrow \text{Quantile}(\mathcal{R}_{\text{hi}}, \frac{\lceil (1-\alpha_{\text{hi}})(n+1) \rceil}{n})$ .
13:  return  $\hat{\beta}_{\text{lo}}, \hat{\beta}_{\text{hi}}$ 
14: end procedure

15: procedure INFERENCE( $x_{n+1}, \text{model}, h, \hat{\beta}_{\text{lo}}, \hat{\beta}_{\text{hi}}, \dots$ )
16:   Define representation  $\hat{z}_{n+1}$  and direction  $\Delta_{n+1}$ .
17:    $S(x_{n+1}) \leftarrow [h(g(\hat{z}_{n+1} - \hat{\beta}_{\text{lo}}\Delta_{n+1})), h(g(\hat{z}_{n+1} + \hat{\beta}_{\text{hi}}\Delta_{n+1}))]$ .
18:   return  $S(x_{n+1})$ 
19: end procedure
```

C.3 Score-Finding Procedures

The core computational task in the calibration stage is finding the minimal perturbation magnitude (β) that satisfies the coverage condition. We present two search methods, linear and binary, each with symmetric and asymmetric versions.

C.3.1 Linear Search

With the linear search method, we perform a straightforward, iterative approach to finding the non-conformity score. We start with a perturbation magnitude of zero and incrementally increase it by a fixed step size, β_{step} . At each step, we perform a forward pass to check if the coverage condition is met. We terminate the search as soon as the condition is satisfied, returning the current magnitude as the score. While simple and robust, this method can be computationally expensive.

Algorithm 3 Symmetric Linear Search

```
1: function SYMMETRIC-LINEAR-SEARCH( $\hat{z}, \Delta, y$ )
2:    $\beta \leftarrow 0$ .
3:   repeat
4:      $\beta \leftarrow \beta + \beta_{\text{step}}$ .
5:   until  $y \in P(\hat{z}, \Delta, \beta)$ 
6:   return  $\beta$ .
7: end function
```

Algorithm 4 Asymmetric Linear Search

```
1: function ASYMMETRIC-LINEAR-SEARCH( $\hat{z}, \Delta, y$ )
2:    $\beta_{lo} \leftarrow 0$ . Repeat  $\beta_{lo} \leftarrow \beta_{lo} + \beta_{step}$  until  $y \geq P(\hat{z}, \Delta, \beta_{lo})_{lower}$ .
3:    $R_{lo} \leftarrow \beta_{lo}$ .
4:    $\beta_{hi} \leftarrow 0$ . Repeat  $\beta_{hi} \leftarrow \beta_{hi} + \beta_{step}$  until  $y \leq P(\hat{z}, \Delta, \beta_{hi})_{upper}$ .
5:    $R_{hi} \leftarrow \beta_{hi}$ .
6:   return  $(R_{lo}, R_{hi})$ .
7: end function
```

C.3.2 Binary Search

With the binary search method, we use a highly efficient divide and conquer algorithm. We first define a search interval, $[0, \beta_{range}]$, and repeatedly halve this interval by testing its midpoint. If the midpoint magnitude achieves coverage, we continue the search in the lower half; otherwise, we continue in the upper half. We repeat this process for a fixed number of iterations, k_{max} , allowing the search to converge on the minimal required magnitude with logarithmic time complexity.

Algorithm 5 Symmetric Binary Search

```
1: function SYMMETRIC-BINARY-SEARCH( $\hat{z}, \Delta, y$ )
2:    $b_{low} \leftarrow 0, b_{high} \leftarrow \beta_{range}$ .
3:   for  $k = 1, \dots, k_{max}$  do
4:      $b_{mid} \leftarrow (b_{low} + b_{high})/2$ .
5:     if  $y \in P(\hat{z}, \Delta, b_{mid})$  then  $b_{high} \leftarrow b_{mid}$ 
6:     else  $b_{low} \leftarrow b_{mid}$ 
7:     end if
8:   end for
9:   return  $b_{high}$ .
10: end function
```

Algorithm 6 Asymmetric Binary Search

```
1: function ASYMMETRIC-BINARY-SEARCH( $\hat{z}, \Delta, y$ )
2:   Upper Bound Score ( $R_{hi}$ ):
3:    $b_{low} \leftarrow 0, b_{high} \leftarrow \beta_{range}$ .
4:   for  $k = 1, \dots, k_{max}$  do
5:      $b_{mid} \leftarrow (b_{low} + b_{high})/2$ .
6:     if  $y \leq P(\hat{z}, \Delta, b_{mid})_{upper}$  then  $b_{high} \leftarrow b_{mid}$ 
7:     else  $b_{low} \leftarrow b_{mid}$ 
8:     end if
9:   end for
10:   $R_{hi} \leftarrow b_{high}$ .
11:  Lower Bound Score ( $R_{lo}$ ):
12:   $b_{low} \leftarrow 0, b_{high} \leftarrow \beta_{range}$ .
13:  for  $k = 1, \dots, k_{max}$  do
14:     $b_{mid} \leftarrow (b_{low} + b_{high})/2$ .
15:    if  $y \geq P(\hat{z}, \Delta, b_{mid})_{lower}$  then  $b_{high} \leftarrow b_{mid}$ 
16:    else  $b_{low} \leftarrow b_{mid}$ 
17:    end if
18:  end for
19:   $R_{lo} \leftarrow b_{high}$ .
20:  return  $(R_{lo}, R_{hi})$ .
21: end function
```

C.3.3 Computational Complexity of Search Methods

The efficiency of the calibration stage is dominated by the score-finding function, as it requires repeated forward passes through the model’s decoder.

- **Linear Search:** The computational complexity is $\mathcal{O}(N)$, where $N = \beta_{\max}/\beta_{\text{step}}$ is the number of steps in the search. In the worst case, this requires N expensive forward passes for each calibration sample.
- **Binary Search:** The complexity is $\mathcal{O}(\log(N))$, where the search space is effectively reduced by half in each step. For a fixed number of iterations $k_{\max}$, the number of forward passes is constant and small (e.g., 30-100), regardless of the range of β . This offers a dramatic speedup over the linear search, especially when high precision is required.

D Additional Tables

D.1 Empirical Coverages

Dataset	α	Coverage						
		COMPASS-J	COMPASS-L	E2E-CQR	Local	Output-CQR	SCP	FCP
H&E	0.05	0.954±0.019	0.959±0.02	0.959±0.019	0.954±0.018	0.955±0.017	0.958±0.018	0.952±0.010
	0.1	0.895±0.029	0.9±0.026	0.906±0.027	0.903±0.031	0.902±0.028	0.906±0.027	0.897±0.014
	0.15	0.85±0.028	0.859±0.032	0.852±0.031	0.856±0.031	0.855±0.03	0.859±0.031	0.850±0.018
Skin Lesion	0.05	0.95±0.01	0.95±0.009	0.951±0.01	0.951±0.009	0.949±0.01	0.952±0.01	0.949±0.006
	0.1	0.901±0.014	0.899±0.012	0.899±0.013	0.9±0.013	0.9±0.014	0.9±0.014	0.903±0.007
	0.15	0.85±0.015	0.851±0.015	0.851±0.013	0.852±0.015	0.849±0.015	0.852±0.015	0.850±0.008
Nodules	0.05	0.951±0.017	0.951±0.015	0.953±0.018	0.951±0.018	0.949±0.019	0.955±0.017	0.950±0.010
	0.1	0.899±0.026	0.902±0.024	0.906±0.026	0.902±0.026	0.901±0.022	0.907±0.027	0.899±0.014
	0.15	0.85±0.032	0.85±0.033	0.857±0.033	0.848±0.031	0.848±0.032	0.852±0.027	0.856±0.016
PolyP	0.05	0.952±0.021	0.951±0.021	0.954±0.021	0.951±0.02	0.95±0.021	0.953±0.02	0.952±0.013
	0.1	0.899±0.032	0.899±0.03	0.907±0.026	0.907±0.027	0.903±0.029	0.905±0.029	0.901±0.017
	0.15	0.85±0.031	0.852±0.032	0.857±0.033	0.853±0.035	0.852±0.034	0.858±0.036	0.854±0.02

Table 2: **Empirical coverage for U-Net.** All methods reach (close to) target coverage.

D.2 COMPASS versus FCP

Dataset	α	Interval Size (Mean±STD, Pixels ²)		
		COMPASS-J	COMPASS-L	FCP
H&E	0.05	4635±569	4542±449	9702±1535
	0.1	2512±160	2655±205	6463±1193
	0.15	2364±145	2513±145	4979±647
Skin Lesion	0.05	1590±61	1633±84	14838±10
	0.1	1149±41	1170±47	12445±51
	0.15	920±24	932±26	8267±79
Nodules	0.05	3184±190	3293±208	16247±2
	0.1	2388±130	2469±141	16143±3
	0.15	1940±106	2005±138	16056±4
PolyP	0.05	5061±497	5358±563	15937±9
	0.1	3940±262	4101±286	15644±10
	0.15	3228±191	3414±254	15353±14

Table 3: **Comparison between COMPASS and FCP interval lengths.** Because FCP fails to reliably converge when finding a non-conformity score, we instead provide a comparison, where we empirically find the minimal radius of an $\mathcal{L}_p$ ball that achieves the target coverage across the test set. COMPASS methods consistently achieve the most efficient interval lengths.

D.3 Weighted Conformal Prediction Interval Size and Coverage

Dataset	Weighting	Method	Interval Size (Mean $\pm$ STD, Pixels ²)	Coverage (Mean $\pm$ STD)
H&E	Class	COMPASS-J	2055.0 $\pm$ 220.0	0.911 $\pm$ 0.024
		COMPASS-L	2065.0 $\pm$ 263.0	0.903 $\pm$ 0.024
		E2E-CQR	2444.0 $\pm$ 278.0	0.9 $\pm$ 0.027
		Local	3113.0 $\pm$ 201.0	0.907 $\pm$ 0.024
		Output-CQR	2890.0 $\pm$ 265.0	0.901 $\pm$ 0.024
		SCP	2371.0 $\pm$ 290.0	0.9 $\pm$ 0.026
	Feature	COMPASS-J	1916.0 $\pm$ 177.0	0.909 $\pm$ 0.024
		COMPASS-L	1944.0 $\pm$ 244.0	0.896 $\pm$ 0.024
		E2E-CQR	2197.0 $\pm$ 288.0	0.884 $\pm$ 0.029
	Jacobian	COMPASS-J	1822.0 $\pm$ 200.0	0.896 $\pm$ 0.031
	Unweighted	COMPASS-J	1810.0 $\pm$ 127.0	0.886 $\pm$ 0.025
		COMPASS-L	1759.0 $\pm$ 159.0	0.884 $\pm$ 0.024
		E2E-CQR	1993.0 $\pm$ 196.0	0.871 $\pm$ 0.026
		Local	2833.0 $\pm$ 158.0	0.88 $\pm$ 0.027
		Output-CQR	2563.0 $\pm$ 134.0	0.879 $\pm$ 0.024
		SCP	2003.0 $\pm$ 169.0	0.878 $\pm$ 0.024
Skin Lesion	Class	COMPASS-J	1318.0 $\pm$ 132.0	0.896 $\pm$ 0.019
		COMPASS-L	1373.0 $\pm$ 136.0	0.895 $\pm$ 0.019
		E2E-CQR	1335.0 $\pm$ 277.0	0.887 $\pm$ 0.021
		Local	2426.0 $\pm$ 354.0	0.893 $\pm$ 0.02
		Output-CQR	3380.0 $\pm$ 43.0	0.888 $\pm$ 0.018
		SCP	1905.0 $\pm$ 475.0	0.888 $\pm$ 0.024
	Feature	COMPASS-J	1573.0 $\pm$ 158.0	0.919 $\pm$ 0.014
		COMPASS-L	1669.0 $\pm$ 168.0	0.919 $\pm$ 0.013
		E2E-CQR	2030.0 $\pm$ 269.0	0.919 $\pm$ 0.012
	Jacobian	COMPASS-J	1485.0 $\pm$ 126.0	0.914 $\pm$ 0.013
	Unweighted	COMPASS-J	2183.0 $\pm$ 132.0	0.94 $\pm$ 0.005
		COMPASS-L	2211.0 $\pm$ 150.0	0.939 $\pm$ 0.007
		E2E-CQR	3400.0 $\pm$ 226.0	0.945 $\pm$ 0.006
		Local	5446.0 $\pm$ 380.0	0.947 $\pm$ 0.006
		Output-CQR	4201.0 $\pm$ 145.0	0.945 $\pm$ 0.008
		SCP	4707.0 $\pm$ 323.0	0.948 $\pm$ 0.006

Table 4: **Comparison between baseline and COMPASS methods for Weighted CP.** For $\alpha = 0.1$, weighted COMPASS methods consistently outperform baseline methods in terms of interval length and restoring coverage under covariate shift.

D.4 Other architectures Interval Size and Coverage

Dataset	α	Interval Size (Mean $\pm$ STD, Pixels ²)					
		COMPASS-J	COMPASS-L	E2E-CQR	Local	Output-CQR	SCP
H&E	0.05	6129 $\pm$ 1147	4680$\pm$595	5115 $\pm$ 541	7649 $\pm$ 821	6860 $\pm$ 571	6021 $\pm$ 858
	0.1	3283 $\pm$ 262	3217$\pm$247	3575 $\pm$ 344	4800 $\pm$ 700	4509 $\pm$ 389	3911 $\pm$ 371
	0.15	2483 $\pm$ 226	2400$\pm$244	2563 $\pm$ 182	3491 $\pm$ 333	3358 $\pm$ 285	2783 $\pm$ 317
PolyP	0.05	5890$\pm$417	6068 $\pm$ 435	8495 $\pm$ 1012	7992 $\pm$ 1324	8296 $\pm$ 822	7774 $\pm$ 930
	0.1	4507$\pm$529	4737 $\pm$ 486	4941 $\pm$ 949	4837 $\pm$ 611	5061 $\pm$ 599	5169 $\pm$ 559
	0.15	3188 $\pm$ 500	3569 $\pm$ 572	2668$\pm$255	3594 $\pm$ 326	3556 $\pm$ 401	3467 $\pm$ 389
Skin Lesion	0.05	1599$\pm$97	1829 $\pm$ 191	2281 $\pm$ 146	4034 $\pm$ 256	10942 $\pm$ 53	2987 $\pm$ 227
	0.1	1108$\pm$45	1146 $\pm$ 45	1236 $\pm$ 66	2311 $\pm$ 167	3056 $\pm$ 45	1685 $\pm$ 135
	0.15	873 $\pm$ 27	927 $\pm$ 27	857$\pm$52	1710 $\pm$ 64	2293 $\pm$ 32	1083 $\pm$ 49
Thyroid Nodules	0.05	3675 $\pm$ 480	3577 $\pm$ 413	4461 $\pm$ 337	3513$\pm$212	6197 $\pm$ 64	4068 $\pm$ 388
	0.1	2671 $\pm$ 251	2645$\pm$216	3390 $\pm$ 197	2762 $\pm$ 129	3942 $\pm$ 102	2826 $\pm$ 200
	0.15	2125$\pm$203	2127 $\pm$ 203	2646 $\pm$ 128	2355 $\pm$ 131	2937 $\pm$ 95	2188 $\pm$ 167

Table 5: **COMPASS produces the most efficient interval lengths on the majority of datasets across target coverages for SegResNet.** For COMPASS-J, we use the final convolutional layer, which is the only layer in the segmentation head. Even without deeper layers, COMPASS still produces efficient intervals.

Dataset	α	Coverage (Mean $\pm$ STD)					
		COMPASS-J	COMPASS-L	E2E-CQR	Local	Output-CQR	SCP
H&E	0.05	0.956 $\pm$ 0.019	0.949 $\pm$ 0.021	0.954 $\pm$ 0.021	0.949 $\pm$ 0.022	0.949 $\pm$ 0.023	0.953 $\pm$ 0.02
	0.1	0.905 $\pm$ 0.029	0.9 $\pm$ 0.027	0.91 $\pm$ 0.027	0.901 $\pm$ 0.028	0.902 $\pm$ 0.028	0.906 $\pm$ 0.027
	0.15	0.858 $\pm$ 0.033	0.852 $\pm$ 0.034	0.856 $\pm$ 0.031	0.854 $\pm$ 0.033	0.854 $\pm$ 0.033	0.856 $\pm$ 0.035
PolyP	0.05	0.954 $\pm$ 0.022	0.948 $\pm$ 0.022	0.954 $\pm$ 0.022	0.949 $\pm$ 0.023	0.95 $\pm$ 0.021	0.952 $\pm$ 0.02
	0.1	0.905 $\pm$ 0.032	0.899 $\pm$ 0.033	0.909 $\pm$ 0.025	0.904 $\pm$ 0.032	0.9 $\pm$ 0.032	0.908 $\pm$ 0.03
	0.15	0.858 $\pm$ 0.035	0.85 $\pm$ 0.035	0.856 $\pm$ 0.031	0.856 $\pm$ 0.031	0.855 $\pm$ 0.037	0.859 $\pm$ 0.033
Skin Lesion	0.05	0.952 $\pm$ 0.011	0.95 $\pm$ 0.01	0.952 $\pm$ 0.01	0.95 $\pm$ 0.011	0.951 $\pm$ 0.01	0.952 $\pm$ 0.01
	0.1	0.903 $\pm$ 0.014	0.901 $\pm$ 0.013	0.9 $\pm$ 0.014	0.902 $\pm$ 0.016	0.901 $\pm$ 0.013	0.902 $\pm$ 0.015
	0.15	0.854 $\pm$ 0.015	0.853 $\pm$ 0.016	0.851 $\pm$ 0.017	0.852 $\pm$ 0.015	0.854 $\pm$ 0.014	0.854 $\pm$ 0.015
Thyroid Nodules	0.05	0.958 $\pm$ 0.017	0.953 $\pm$ 0.017	0.954 $\pm$ 0.019	0.951 $\pm$ 0.019	0.953 $\pm$ 0.017	0.952 $\pm$ 0.018
	0.1	0.902 $\pm$ 0.026	0.901 $\pm$ 0.027	0.909 $\pm$ 0.025	0.9 $\pm$ 0.024	0.901 $\pm$ 0.027	0.904 $\pm$ 0.026
	0.15	0.852 $\pm$ 0.026	0.85 $\pm$ 0.026	0.856 $\pm$ 0.029	0.848 $\pm$ 0.032	0.849 $\pm$ 0.032	0.851 $\pm$ 0.03

Table 6: **Empirical coverage for SegResNet.** All methods reach (close to) target coverage.

E Experimental Details

Architectures. We use the standard U-Net, SegResNet, and SwinUNETR architecture available in the MONAI framework [25]. For the U-Net, we used an encoder with channel sizes of (32, 64, 128, 256) and a corresponding decoder, using two residual units per block, batch normalization, and a dropout rate of 0.1. For SegResNet, we used a residual encoder-decoder network that started with 32 initial filters and included varying numbers of blocks in its down-sampling path (1, 2, 2, 4), batch normalization, and a dropout probability of 0.1. For SwinUNETR, we used a transformer-based model configured for an image size of (128, 128) pixels, with a feature size of 48 and multi-head attention mechanisms across four depth levels.

Preprocessing. To standardize all experiments, we resize each image and segmentation mask to a standard resolution of 128x128 pixels. For data augmentation during training, we apply a set of transformations consisting of a random crop to the specified size, followed by random horizontal and vertical flips, each with a equal probability.

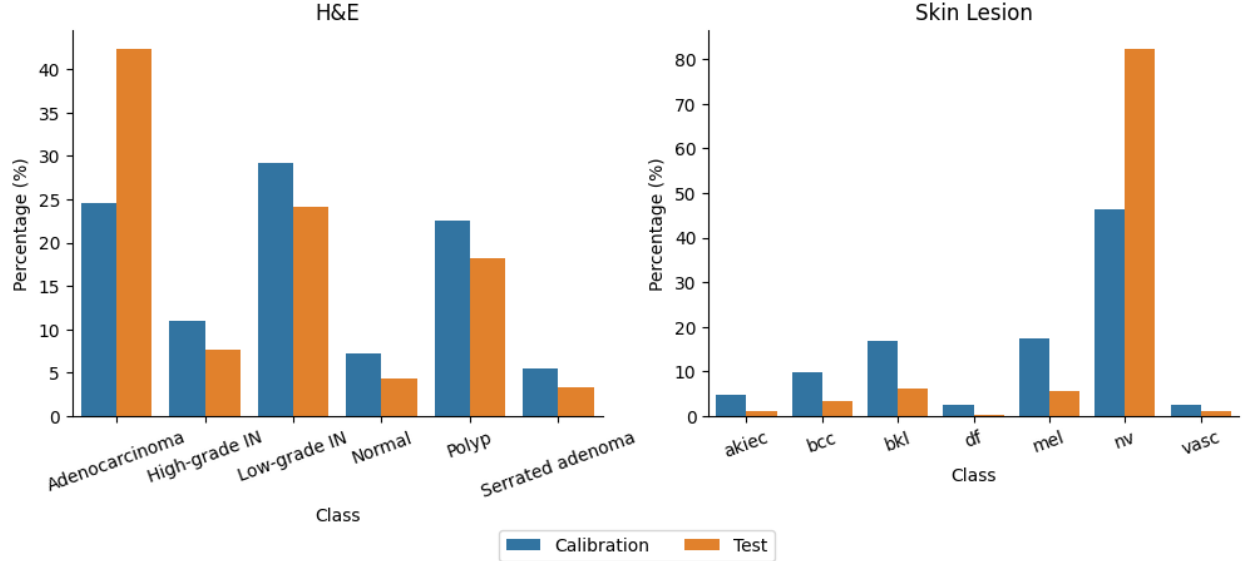


Figure 5: **Adversarial shift distribution for H&E and Skin Lesion datasets.** For H&E, we place 60% and 40% of the Adenocarcinoma (easy) samples into the calibration and test sets. For Skin Lesion, we place 70% and 30% of the melanocytic nevi (hard) samples into the calibration and test sets.

Training details. For each architecture, we trained both a standard model and an End-to-End Conformalized Quantile Regression (E2E-CQR) variant [24], with all models being trained using an AdamW optimizer with a learning rate of $1e-4$ and a batch size of 32. The standard models were configured for binary segmentation tasks, taking a 1 (for grayscale) or 3-channel (for RGB image) as input and producing a single-channel output mask. Training was optimized using a Dice score loss function to maximize the overlap between the predicted and ground-truth segmentations. Following the original E2E-CQR model [24], we modify the number of output channels to 3 to produce three distinct segmentation masks, corresponding to the lower quantile, the median prediction, and the upper quantile, which together form the prediction interval. These models were trained using the Tversky loss [33]. Thus, it allows the model to directly learn the uncertainty bounds end-to-end.

Baseline Calibration Details. We follow the standard frameworks for SCP [8, 29], CQR [30], and Local CP [31, 32, 29]. For CQR, we use two separate Gradient Boosting Regressors to learn the lower and upper quantiles of the prediction and adjust the resulting We use a learning rate of 0.1, 50 estimators, a maximum depth of 3, minimum leaf samples of 1, and a minimum samples to split of 9. For Local CP, we use two separate Random Forest Regressors to learn the mean and mean absolute difference. We use 1000 estimators and a minimum leaf sample of 100. For E2E-CQR, we follow the same calibration procedure as CQR. However, in this case, the quantiles are learned by the model.

COMPASS Calibration and Testing Details. For all experiments, we run both the symmetric and asymmetric versions of COMPASS with 1 component on the logits and each layer in the segmentation head. We report the layer with the minimum mean interval length and its corresponding coverage. We perform 100 random calibration-test splits. For weighted CP, we use LightGBM with the default settings [37]. We also perform 100 calibration-test splits. However, in this case, the samples were shuffled to maintain the target proportion of classes. We show the covariate shift in Figure 5.

F β - δA Trajectories

We verify the nestedness of linear latent perturbations (Definition1). For each dataset, we perform a sweep across different β s and compute the change in volume $\delta A = \frac{A_\beta - A_0}{A_0} \times 100$ where A_β is the area when the original latent is perturbed by β .

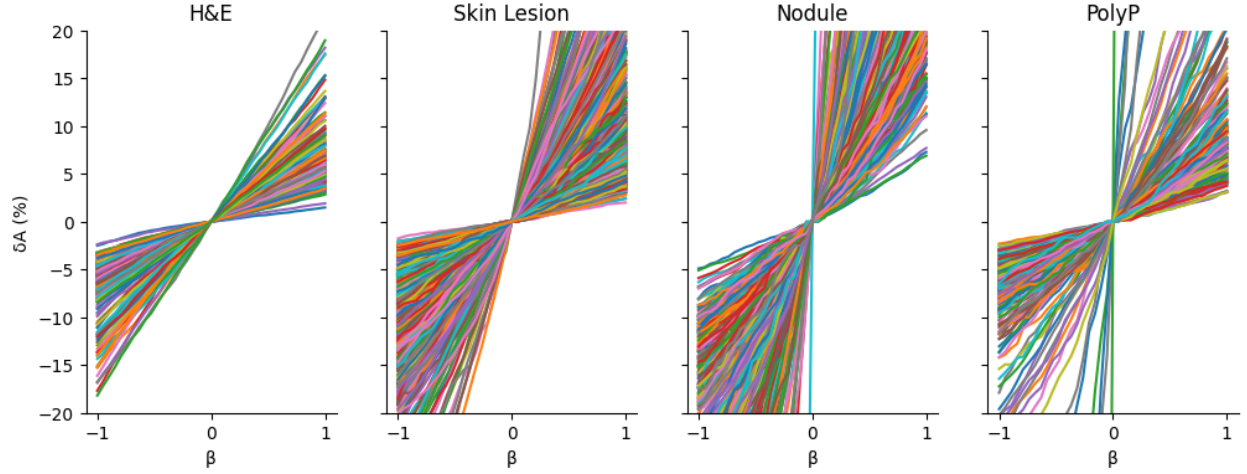


Figure 6: Figure perturbation trajectories for 4 datasets for U-Net

G Additional Figures

G.1 Segmentation overlays for different percentages of change in segmentation area (δA) for COMPASS-J

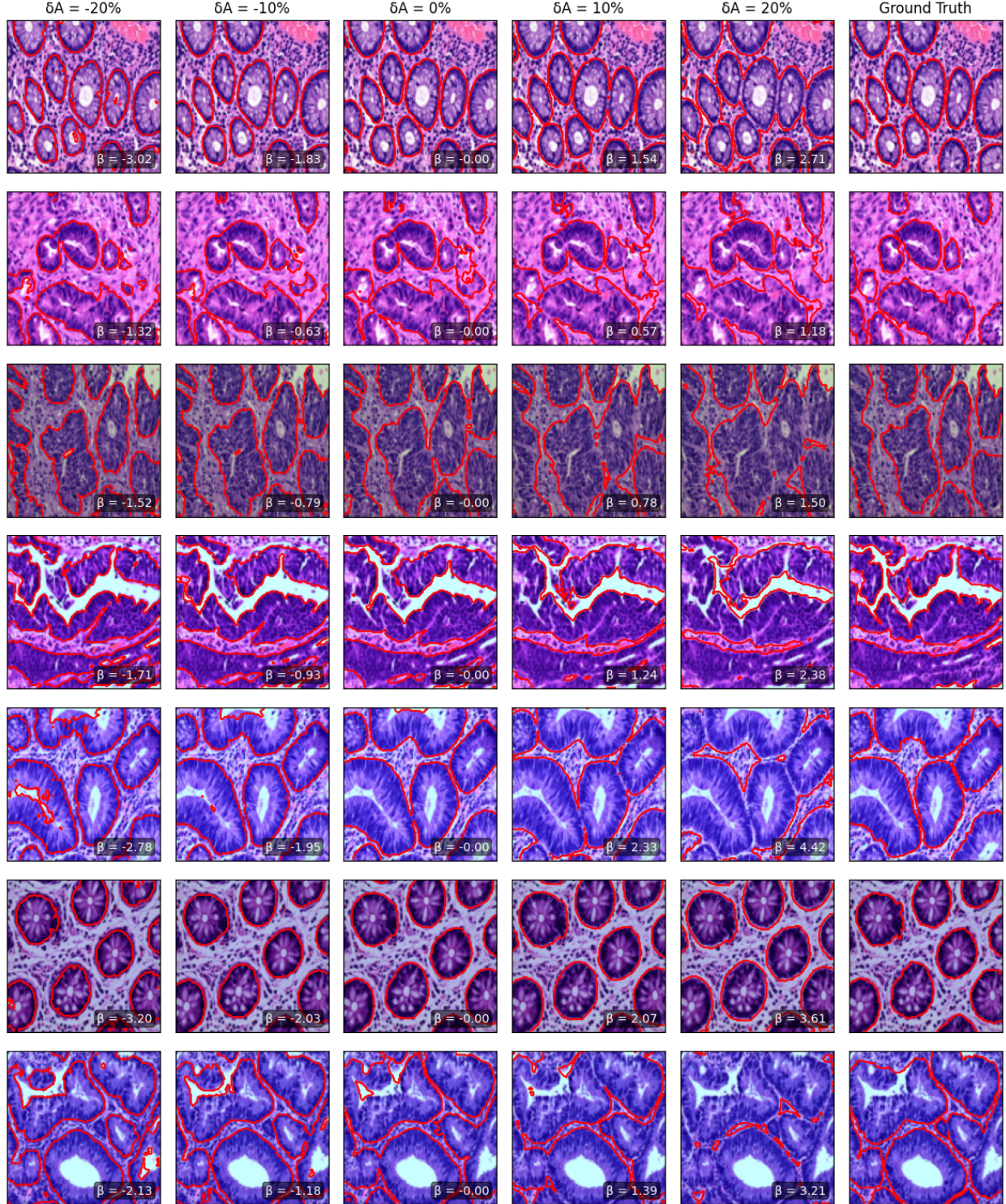


Figure 7: Segmentation volume increases with β for H&E. Each row is a different sample. $\delta A = 0\%$ is the original prediction.

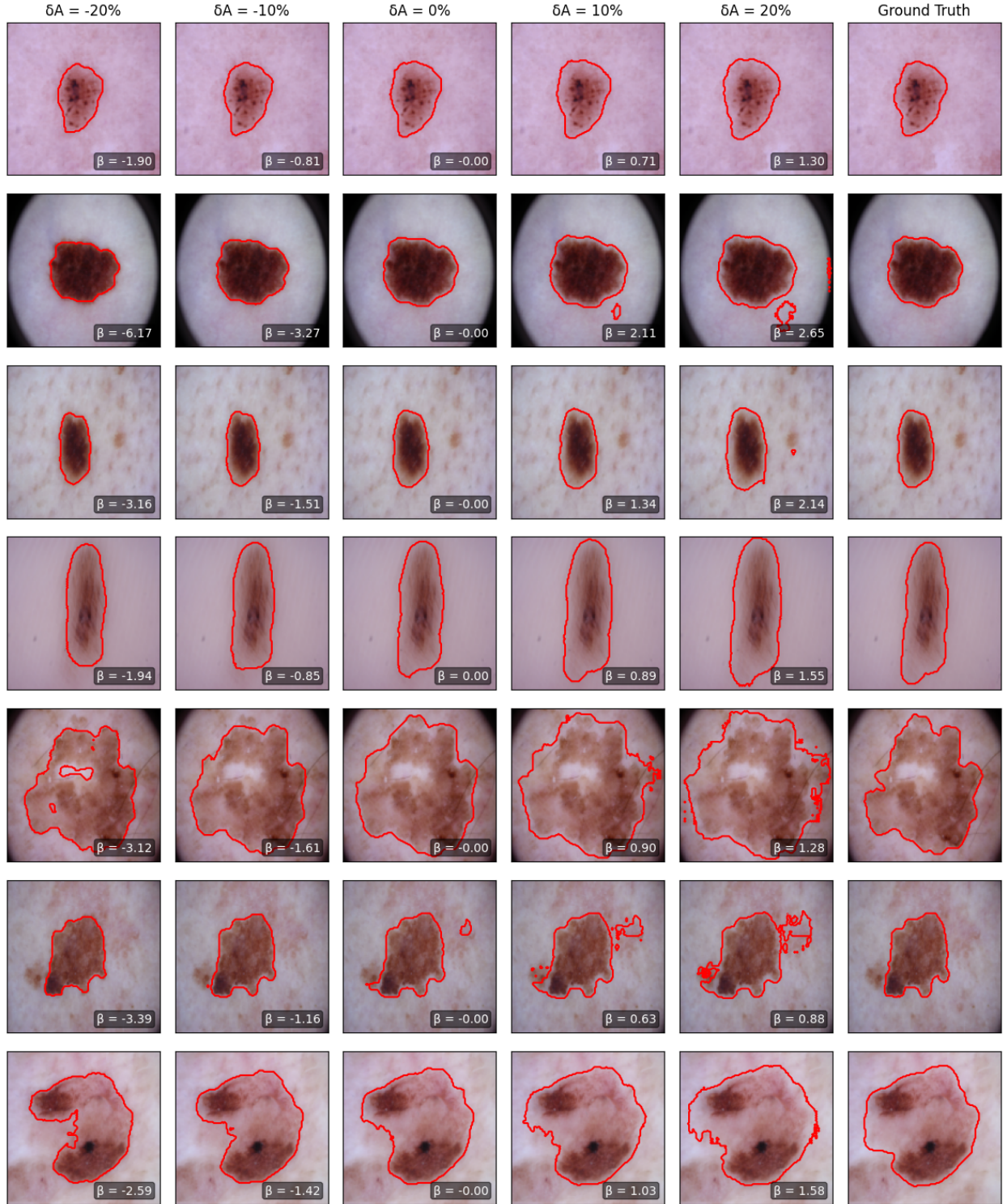


Figure 8: Segmentation volume increases with β for Skin Lesion. Each row is a different sample. $\delta A = 0\%$ is the original prediction.

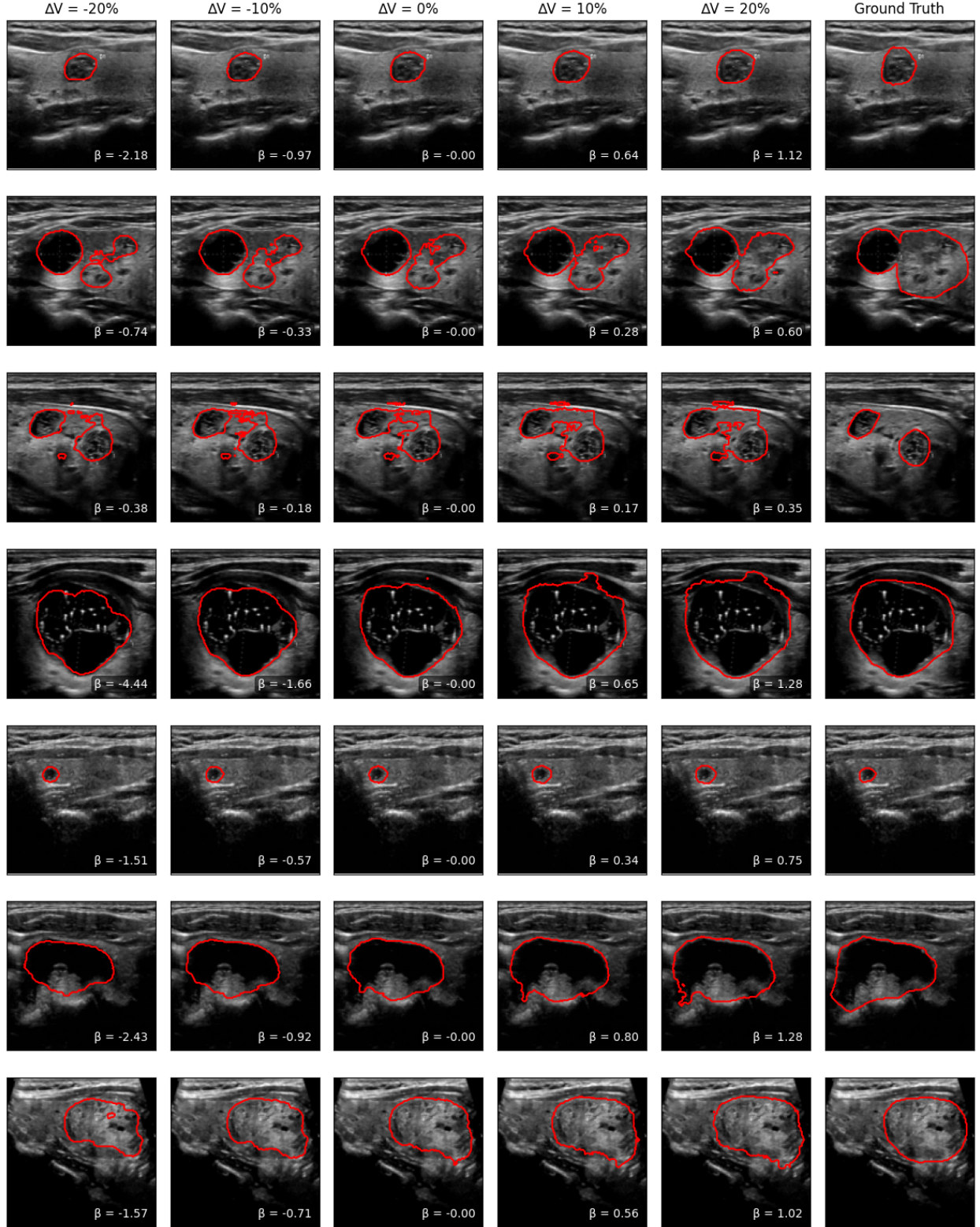


Figure 9: Segmentation volume increases with β for Thyroid Nodule. Each row is a different sample. $\delta A = 0\%$ is the original prediction.

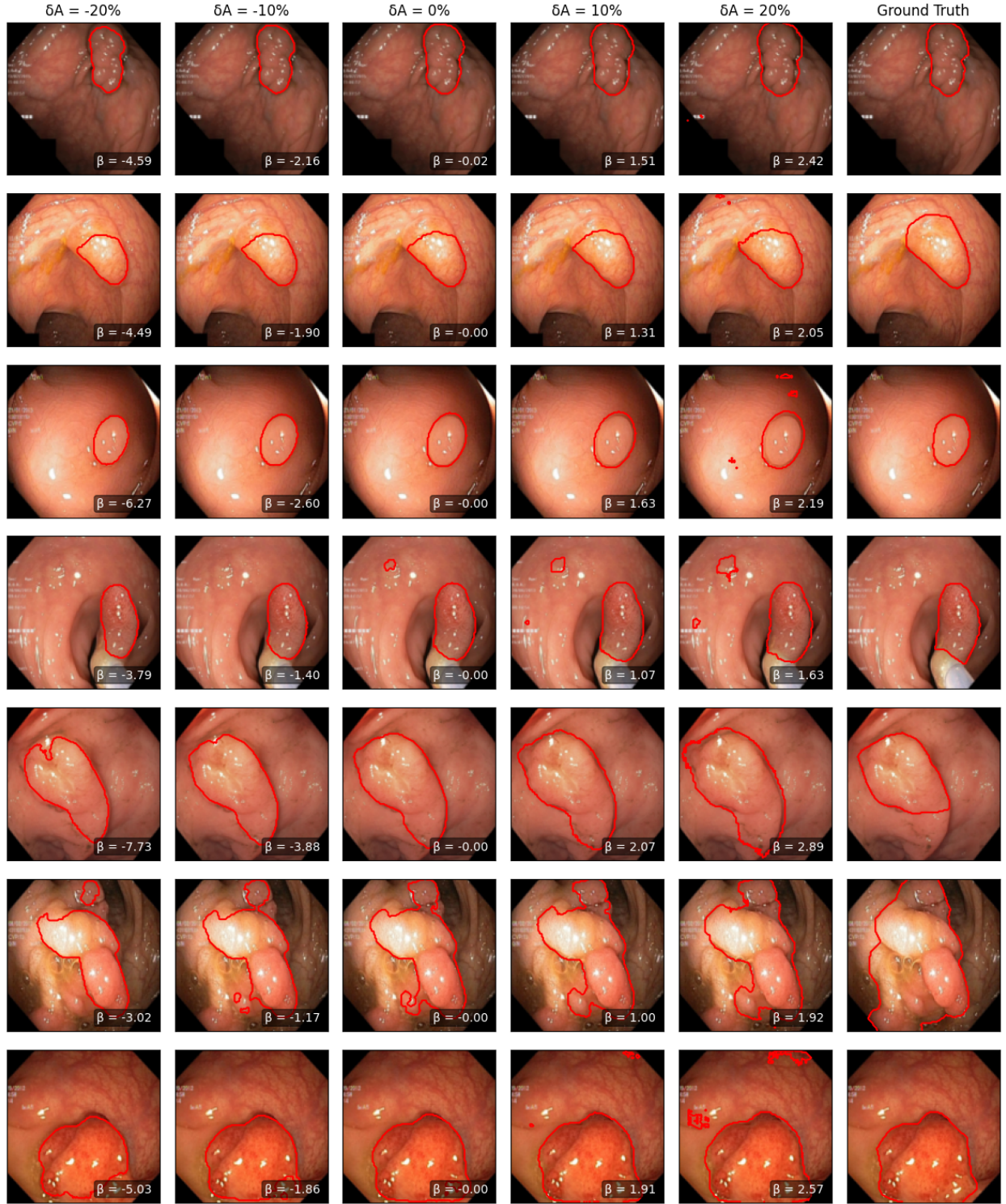


Figure 10: Segmentation volume increases with β for PolyP. Each row is a different sample. $\delta A = 0\%$ is the original prediction.

G.2 Segmentation overlays for different percentages of change in segmentation area (δA) for COMPASS-L

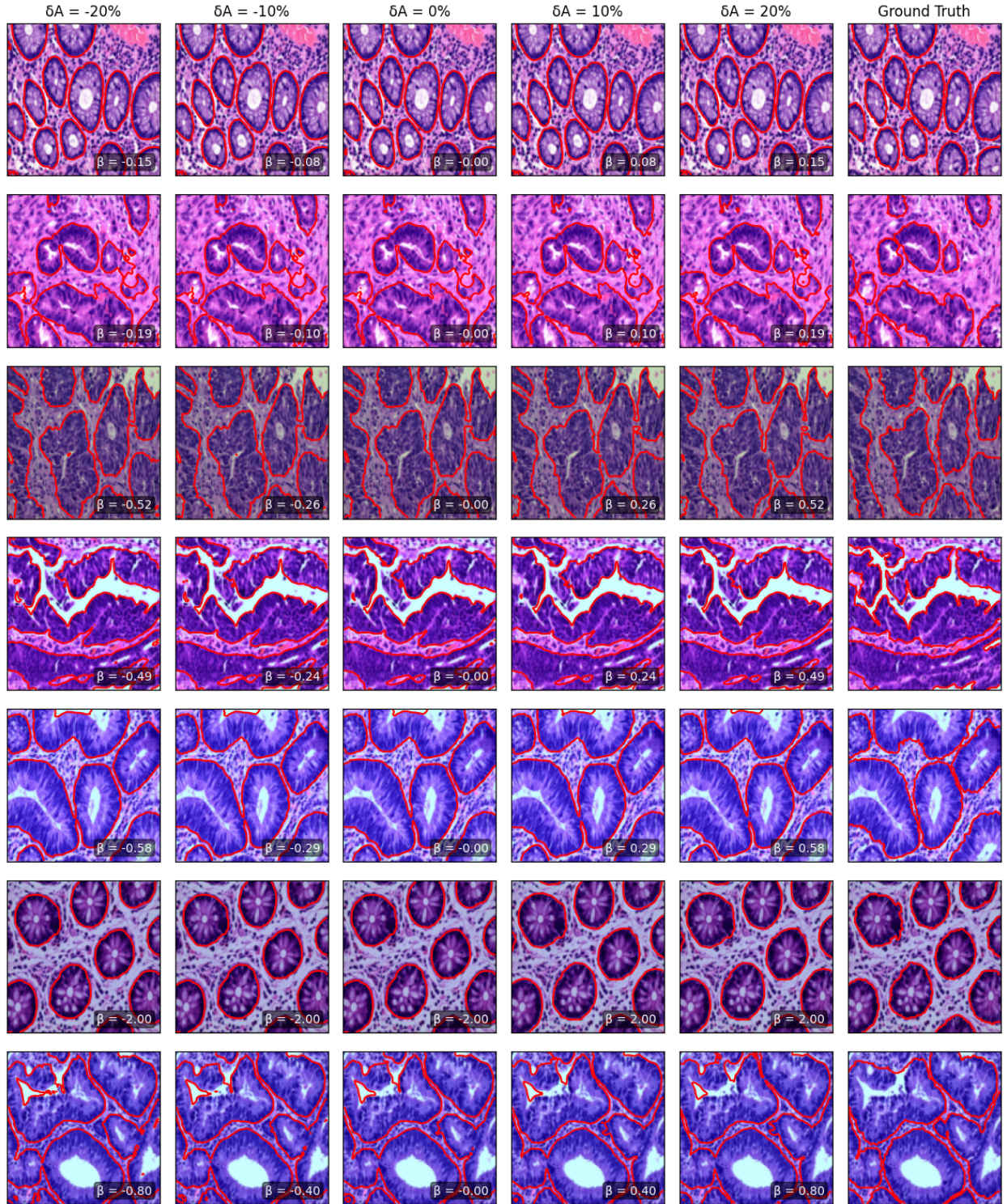


Figure 11: Segmentation volume increases with β for H&E. Each row is a different sample. $\delta A = 0\%$ is the original prediction.

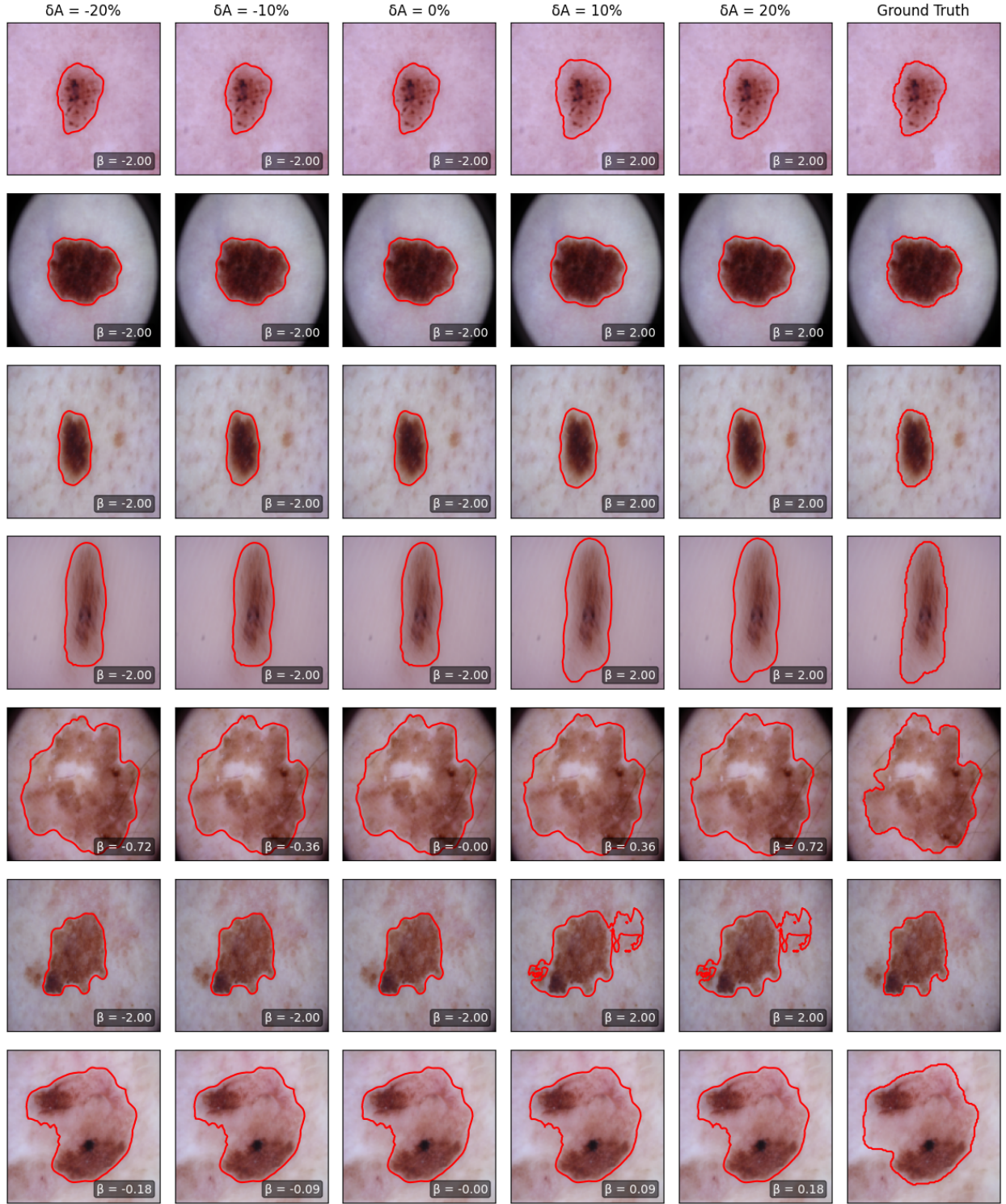


Figure 12: Segmentation volume increases with β for Skin Lesion. Each row is a different sample. $\delta A = 0\%$ is the original prediction.

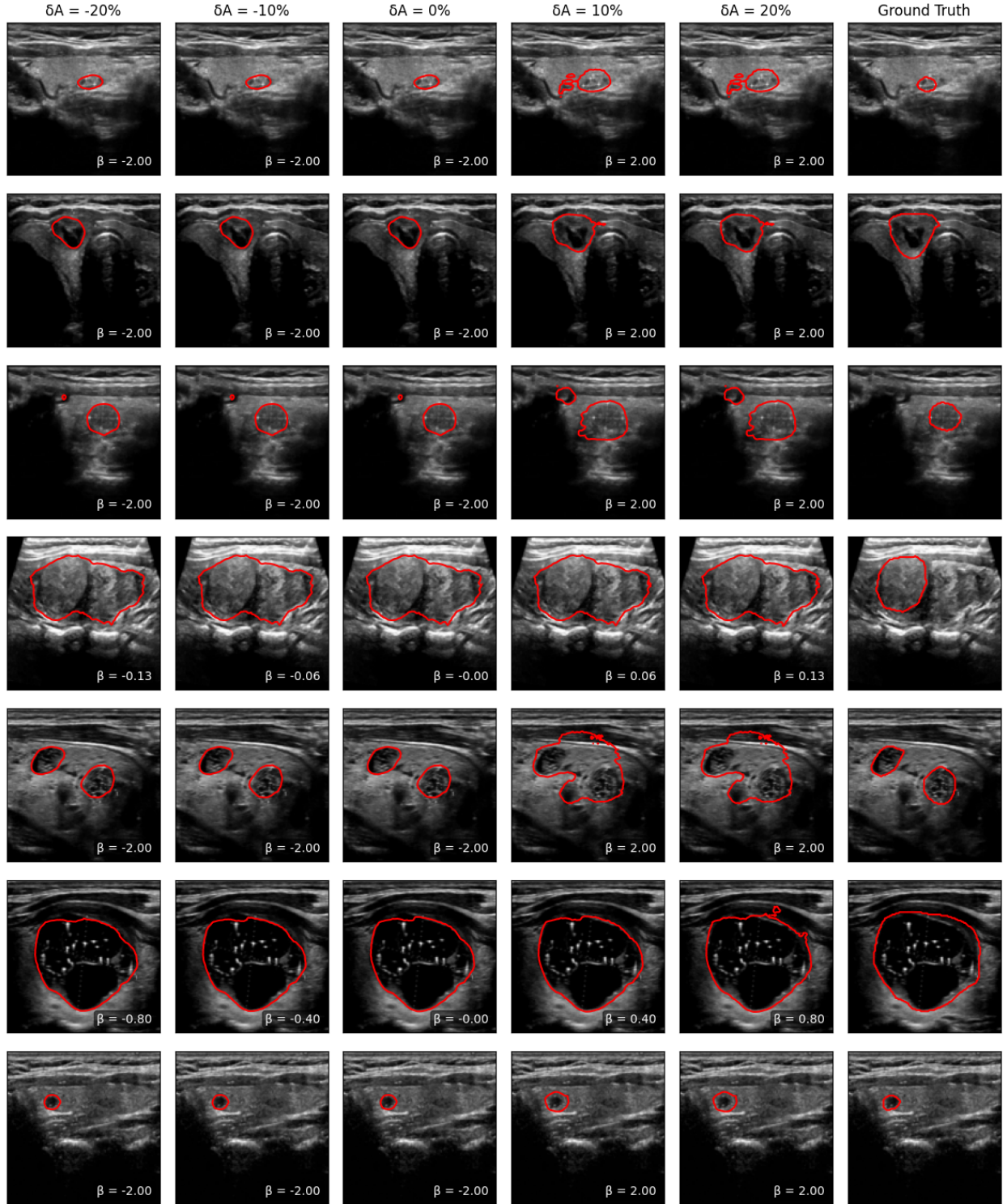


Figure 13: Segmentation volume increases with β for Thyroid Nodule. Each row is a different sample. $\delta A = 0\%$ is the original prediction.

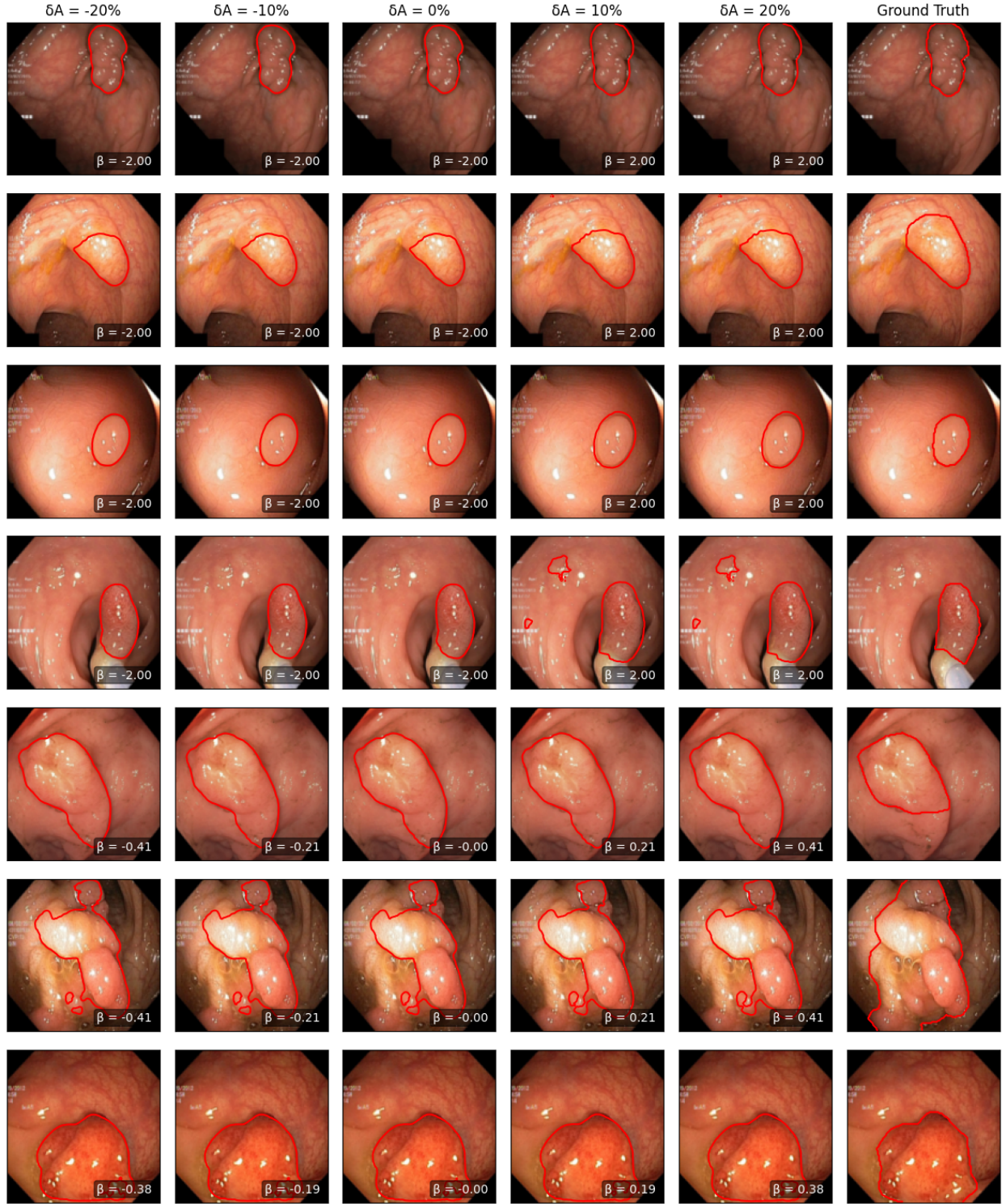


Figure 14: Segmentation volume increases with β for PolyP. Each row is a different sample. $\delta A = 0\%$ is the original prediction.